

# Context Selection Is a Partial Order: A Blackwell Framework and Certified LLM Evidence

Andriy Bilous

Department of Information Systems and Networks  
Lviv Polytechnic National University, Lviv, Ukraine

ORCID: 0009-0006-5467-7932

andriy.bilous@uitware.com

Oleksandr Reminnyi

Department of Computerized Automatic Systems  
Lviv Polytechnic National University, Lviv, Ukraine

ORCID: 0009-0005-5119-3695

## Abstract

Context selection for language models is almost universally posed as a *scalar* ranking problem: score each candidate source, take the top- $k$ . We argue this collapses a partial order. Modeling each context source as a statistical *experiment* on the task’s latent requirement, we transport Blackwell’s comparison of experiments [Blackwell, 1953] to context selection: when two sources are Blackwell-incomparable, no query-independent scalar can rank them correctly across all tasks, a formal account of the field’s “relevance-does-not-transfer” folklore. To make the order measurable on real models behind executable verifiers, we define a decision-restricted value-deficiency *surrogate* and equip it with a distribution-free finite-sample *certificate*. On six language models across four vendors we certify ( $\geq 90\%$  confidence, both directions) that two private context sources are empirically incomparable. Representative lexical, dense bi-encoder, and cross-encoder rankers all prefer the wrong source on a confound-controlled trap, while an LLM listwise reranker that reads the decisive content escapes, the topicality boundary the theory predicts. We also show that supplying a strict *superset* of a sufficient source collapses certified pass rates from 100% to as low as 0%, with order, length-padding, and routing-instruction controls attributing the harm to the added signal, not position or length. The conclusion is methodological: context selection is a partial-order problem, and the right operational object is a certified deficiency surrogate, not a scalar. The constructive selection rule this implies is developed in a companion paper; scaling certified selection to corpus retrieval is the central problem we leave open.

## 1 Introduction

Suppose a coding assistant is handed one of two short notes about a software module. The first note says how to *call* its main function: the order of the arguments, a keyword that is mandatory, and the error it raises on failure. The second note says how the module *encodes* numbers: a digit format ending in a checksum. For a task that calls the function the first note is decisive and the second is useless; for a task that encodes a number it is the reverse; and neither note can be reconstructed from the other. This everyday situation is the entire paper in miniature. It has a precise name, the two notes are *incomparable* in the sense of Blackwell’s comparison of experiments [Blackwell, 1953], and a precise consequence: no single relevance score can rank the two notes so that the ranking is

right for every task, because a score must crown one winner and here neither note wins outright. The body makes this exact, gives a statistical test that *certifies* it from ordinary pass/fail runs on a live model, and finds that six production language models across four vendors all exhibit it. A reader who keeps these two notes in mind can read every symbol below as a statement about the first note versus the second (in the body they are the sources  $W_1$  and  $W_2$ ).

Production language-model systems spend much of their budget deciding *what context to place in the prompt*. Retrieval-augmented generation ranks passages, in-context learning selects demonstrations, and agentic coding tools choose which files and conventions to surface. Across all of these, the operative tool is a **scalar score**, lexical or embedding relevance, mutual information,  $\mathcal{V}$ -usable information [Ethayarajh et al., 2022], expected information gain, used to impose a total order and retain the top candidates. The scalar is convenient: it sorts, thresholds, and composes. We argue it is also the wrong abstraction, and that a classical apparatus from statistical decision theory says precisely why.

**A folklore observation, restated as structure.** A persistent, informal observation undercuts the scalar paradigm: *relevance does not transfer across tasks*. A source that helps one task hurts another; “the best context” is task-dependent. This is usually treated as a modeling nuisance to be engineered around with better rankers. We show it is the signature of a *partial order*. We model a context source  $W$  as a statistical experiment on the task’s latent requirement  $S$  (the object a verifier checks; §3), which lets us import two classical tools. Blackwell’s order ranks experiments by garbling:  $W_1 \succeq_B W_2$  iff  $W_2$ ’s signal is a noised (garbled) version of  $W_1$ ’s, and, by the Blackwell–Sherman–Stein theorem, this holds *iff* an idealized decision-maker does at least as well with  $W_1$  as with  $W_2$  on *every* decision problem [Blackwell, 1953, Torgersen, 1991]. The order is *partial*: many source pairs are incomparable, neither garbling the other. Incomparability is exactly the regime in which no scalar summary can be faithful, because a scalar is a total order and a total order extending a strict partial order must invert at least one incomparable pair on some task.

**What is and is not new.** The machinery, the garbling order, the Blackwell–Sherman–Stein equivalence, Le Cam’s deficiency, is classical; our contribution is the application, the incomparability framing, and a certified estimator (§2). One gap the framing must respect, and that we close, concerns the decision rule: Blackwell’s theorem is about the *optimal* (Bayes) use of an experiment, whereas every quantity we measure,  $\text{PASS}_M(W, T) = \mathbb{E}[v_T(\delta_M(T, w))]$ , is evaluated at a single fixed sub-optimal rule  $\delta_M$ . We therefore develop the order in a  $\delta_M$ -restricted register with a one-way bridge to the idealized order (§4), confine every empirical claim to the restricted object, and name the estimator a decision-restricted *value-deficiency surrogate*, not a Le Cam deficiency (§6).

**A latent-coordinate model that makes the order non-vacuous.** Fixed strings would naively be Blackwell-trivial constant experiments. We model the latent requirement as a vector of coordinates and each source as a deterministic experiment observing a *projection*; comparison reduces to refinement of the projections’  $\sigma$ -algebras, and incomparability is mutual non-refinement, a property we establish for our concrete pair (Lemma 1).

**Empirical findings.** We instantiate the theory on two private, equally on-topic conventions about the same module: an API call contract  $W_1$  and a wire-encoding invariant  $W_2$  (§7), with a four-cell verifier battery using executable PASS/FAIL judgments (return code 0 is PASS). This imposes a restriction: the battery is a fixed, adversarially constructed opposition, not a sampled task distribution, and the certified incomparability is carried by one API cell and one encoding cell;

we report the effective support explicitly and do not call  $\mathcal{D}$  a “distribution.” On this instrument we obtain, on *real* model calls with no modeled numbers:

- **Certified empirical incomparability on all six models.** The uniform certificate clears  $\geq 90\%$  confidence in *both* directions: the two-sided lower bounds  $L_{\mathcal{D}}(W_1, W_2) / L_{\mathcal{D}}(W_2, W_1)$  are Haiku-4.5 +0.88 / +0.83 ( $n=80$ ), Sonnet-4.6 +0.76 / +0.76 ( $n=40$ ), Opus-4.8 +0.55 / +0.55 ( $n=20$ ), GPT-5.5 +0.76 / +0.76 ( $n=40$ ), DeepSeek-V4-Pro +0.347 / +0.715 ( $n=40$ ), and Kimi-K2.6 +0.675 / +0.762 ( $n=40$ ). Incomparability is therefore not a single-lab artifact.
- **Representative topical rankers mis-rank the pair.** On a trap task saturated with encoding vocabulary, query-conditioned lexical relevance scores  $\varphi(W_1)=0.36 < \varphi(W_2)=0.44$  (ranking  $W_2 \succ W_1$ ), yet realized PASS ranks  $W_1 \gg W_2$  (e.g. 96–100% vs. 0%), and the same mis-rank is realized by two dense bi-encoders and a cross-encoder reranker, while an LLM listwise reranker ranks the trap correctly (§7.4.2). On the honest tasks every ranker agrees with PASS; failure occurs *only* where surface relevance diverges from verifier-decisive content, exactly the boundary Proposition 2 predicts: topical feature-based rankers must fail there, while rankers that read the decisive coordinate can escape.
- **Certified anti-monotonicity.** The superset  $W_{1+} = W_2 ++ W_1$ , which literally *contains*  $W_1$ ’s signal, sharply degrades the API cells: Sonnet and Kimi collapse 100%  $\rightarrow$  0% (interference  $\Psi_T = -1.00$ , certified upper bound  $\leq -0.69$ ); GPT-5.5 certifies  $\Psi_T \leq -0.60$  on `api_post_ok`. The interference certificate clears on at least one cell for five of six models (Opus reproduces,  $\Psi_T = -0.75 / -0.80$ , but  $n=20$  is too few to certify; we give the power argument in §7). An order control shows the harm does not recover under reordering, so it is not a lost-in-the-middle artifact, and a length-matched padding control shows equal-length irrelevant text is harmless, so it is not a token-count artifact either. One caveat: the effect is large and robust across the capability ladder but is *not* monotone in capability (Sonnet, the middle model, is the starkest; Opus, the strongest, less so), so we claim no capability trend.
- **A supporting observation: private-context value is model-relative.** The none-arm baselines span 0%–75% guessability across vendors (§7.4.4), an empirical instance of the theory’s  $\theta$ -dependence; we report it as supporting evidence that caps which cells can be certified per model, not as a headline claim.

**Methodological asymmetry.** The Anthropic models run agentially (`claude -p`, up to six turns); the non-Anthropic models run single-shot (completion plus code extraction via an OpenAI-style API). We flag this throughout: vendor is confounded with harness, so every *cross-vendor magnitude* comparison (the value spectrum, the interference ladder) inherits that confound, whereas the binary within-model claims do not. The single-shot setting is arguably a *cleaner* probe of whether a model uses supplied context, since it removes the agentic-exploration confound; we treat the asymmetry as a limitation to be reported, not explained away.

**Contributions.** The paper makes one claim, context selection is a partial-order problem rather than a scalar-ranking problem, and contributes exactly the three pieces that claim needs.

- (C1) **Theory: context sources form a Blackwell partial order** (§4, §5). We formalize context sources as statistical experiments on the task’s latent requirement, instantiate the Blackwell garbling order with its for-all-tasks dominance guarantee via the Blackwell–Sherman–Stein theorem, define the  $\delta_M$ -restricted order the experiments probe, and prove a one-way bridging

lemma between them. The central consequence is incomparability: Blackwell-incomparable sources cannot be ranked by *any* query-independent scalar  $\varphi(W)$  for all tasks (Theorem 2), with the query-conditioned topical case isolated separately (Proposition 2). Information retrieval has long argued that no single fixed document score is optimal once documents interact (§2.1); we derive the impossibility as a theorem in the Blackwell frame.

- (C2) **Certificate: a measurable deficiency surrogate with finite-sample guarantees** (§6). We define  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = \max(0, \sup_{T \in \mathcal{D}} [\text{PASS}_M(W_2, T) - \text{PASS}_M(W_1, T)])$ , a decision-restricted value-deficiency surrogate (named as such, not as a Le Cam deficiency), and a distribution-free uniform certificate: exact Clopper–Pearson intervals [Clopper and Pearson, 1934] at Bonferroni level [Bonferroni, 1936]  $\alpha = \eta/(2|\mathcal{D}|)$ , valid jointly on a single event, so that  $L_{\mathcal{D}} > 0$  certifies a positive value gap at confidence  $\geq 1 - \eta$ ; we argue why a naive DKW bound [Dvoretzky et al., 1956] under-covers the supremum, and give the matching interference certificate. This is the instrument that makes the partial order testable on a real model behind a real verifier.
- (C3) **Evidence: certified incomparability and scalar-ranking failure on live models** (§7). Certified empirical incomparability on six models across four vendors; the ranker-spectrum mis-rank on a confound-controlled trap (lexical cosine, dense bi-encoders, and a cross-encoder all fail; an LLM listwise reranker that reads the decisive content escapes, exactly the topicality boundary the theory draws); and certified anti-monotonicity, a strict superset of a sufficient source collapsing PASS, with order, length-padding, and routing-instruction controls, all with the scope limits stated in §8.5.

**Roadmap.** §2 positions the work against scalar context-selection criteria and the recent transport of comparison-of-experiments into machine learning, and frames our delta as application-plus-framing-plus-estimator rather than new core mathematics. §3 gives the latent-coordinate experiment model and defines  $\theta$ . §4 develops the partial order, the  $\delta_M$ -restricted order, the non-refinement construction, and the bridging lemma, and §5 proves the source-level incomparability statement and isolates the query-conditioned case (C1). §6 defines the surrogate, the uniform certificate, and the interference certificate, and relates the surrogate to the true Le Cam deficiency (C2). §7 reports the six-model program with full per-cell tables (C3). §8.5 states threats to validity and scope, and §9 concludes. Proofs of the two central impossibility results (Theorem 2, Proposition 2) accompany their statements; the remaining proofs and technical material are deferred to Appendix A, the extended experimental record to Appendix B, and protocol and artifact details to Appendix C. Every reference has been verified against arXiv/DBLP/publisher records.

## 2 Related Work

**Positioning.** The machinery we use, Blackwell’s garbling order [Blackwell, 1953] and Le Cam’s deficiency [Le Cam, 1964, 1986], is classical, and we claim no new core mathematics. Our contribution is an *application*, a *framing*, and an *estimator*: (i) we treat each candidate context source for a language model as a statistical experiment on the task’s latent requirement and rank sources by the Blackwell order, with a decision-independent, for-all-tasks dominance guarantee (§4); (ii) we observe, as an application-level corollary of Blackwell’s theorem, that Blackwell-*incomparable* sources cannot be ranked by any scalar score for all tasks (§5), which converts the empirical folklore that “relevance does not transfer across tasks” into a theorem, formalizing within the Blackwell frame a limitation that information retrieval has long recognized for interacting documents (§2.1); and (iii) we supply a

Le Cam-inspired, decision-restricted value-deficiency *surrogate* with a distribution-free finite-sample certificate that is auditable on a real model with a real verifier (§6). To our knowledge, no prior work applies the comparison of experiments to selecting inference-time context for language models.

## 2.1 Scalar selection and its limits in information retrieval

Context and demonstration selection is dominated by *scalar* criteria, each inducing a total order and a top- $k$  prefix: topical relevance, lexical or learned [Yu et al., 2024];  $\mathcal{V}$ -usable information [Ethayarajh et al., 2022, Yuan et al., 2025]; expected information gain and Bayesian experimental design [Choudhury et al., 2025, Chan et al., 2025]; game-theoretic and influence-based demonstration valuation [Xie et al., 2024, Vinay et al., 2024]; and binary sufficient-context detection [Joren et al., 2024]. Each collapses the comparison of two sources to one real number and is therefore a scalar of the kind the paper rules out (Theorem 2 for query-independent scores, Proposition 2 for topical query-conditioned ones): not a competitor to out-perform but an instance of the excluded class (we prove Blackwell-monotonicity for the closest neighbour,  $\mathcal{V}$ -usable information, in Proposition 1, and conjecture it for the rest). Two families sit at the boundary of the class. Listwise LLM rerankers [Sun et al., 2023] are query-conditioned *set* functions, outside the theorem’s scope, though they carry no dominance guarantee of any kind; and diversity-aware selection, from MMR [Carbonell and Goldstein, 1998] to DPPs [Ye et al., 2023] and submodular coverage [Huang, 2025], optimizes set objectives over embedding geometry, whose partial-order-grounded analogue (certified coverage of latent coordinates) we develop in a companion paper.

The limits of scalar relevance are one of the oldest themes in information retrieval, and we position inside that lineage rather than apart from it: the Probability Ranking Principle [Robertson, 1977] is optimal only under an independence assumption, its failure once documents interact is long documented [Fuhr, 2008, Carbonell and Goldstein, 1998], and the LLM-era version, relevance-maximizing retrieval degrading answer quality, is measured directly [LeVine and Varjavand, 2025, Qiu et al., 2026]. Our delta is the *form* of the statement: the failure becomes a theorem in the comparison-of-experiments frame, with the precise structural condition, Blackwell incomparability, under which every query-independent scalar must mis-rank, and a certified live instance on six production models.

## 2.2 Task-dependence and context degradation

That added context can degrade performance is by now well documented: a single irrelevant sentence tanks reasoning [Shi et al., 2023]; position matters [Liu et al., 2024]; high-scoring distractors hurt while random noise can help [Cuconasu et al., 2024]; retrieval curves are inverted-U in passage count [Jin et al., 2024]; document count hurts at fixed length [Levy et al., 2025]; added length alone hurts [Du et al., 2025]; and the effect persists across model generations [Hong et al., 2025] and agent-scaffold components [Liu, 2026]. We do not claim to discover this genus. Our anti-monotonicity finding contributes its order-theoretic form: a strict *superset* of a sufficient source, on sources separately *certified* Blackwell-incomparable, collapses PASS under a per-task interference certificate with position, length, and routing controls; the algebraic object  $\Psi_T < 0$  is one no monotone-submodular context-value scalar can carry. Pareto-incomparability of prompts along competing metrics *within* a single task [Zhao et al., 2025] is orthogonal to our object, incomparable *sources across tasks*.

### 2.3 The comparison of experiments in machine learning

The Blackwell–Le Cam apparatus has entered machine learning, but never for inference-time context. Closest is training-*dataset* valuation: Zheng et al. [2024] rank datasets by Blackwell informativeness and show a pointwise-mutual-information *scalar* is faithful in their Bayes-optimal, white-box regime, an instructive contrast with ours, where the order is genuinely partial for a fixed rule and no query-independent scalar can be faithful (Theorem 2); they provide no incomparability statement or finite-sample certificate. Garbling has been used to harden LLM evaluations [Bradley, 2024]; statistical deficiency to order training *tasks* [Fosse et al., 2025]; a deficiency-induced *directional* dominance appears in transfer learning, without an incomparability statement, formal partial order, or distribution-free certificate, and outside language modeling [Akdemir, 2025]; and comparison-of-experiments ideas touch loss design [van Rooyen and Williamson, 2014]. Blackwell’s theorems have been surveyed for AI [Paxton, 2026], but to our knowledge no prior work states a Blackwell partial order over context sources, derives the no-scalar-ranks consequence for language models, or supplies a certified decision-restricted estimator. The mathematics we rely on is classical: the Blackwell–Sherman–Stein equivalence and deficiency are due to Blackwell and Le Cam [Blackwell, 1953, Le Cam, 1964, 1986], with Torgersen’s monograph as the canonical reference, and the fact that the order is a non-lattice partial order with genuinely incomparable pairs is classical as well [Torgersen, 1991]. Theorem 2 is, mathematically, a short corollary of Blackwell’s theorem applied to a constructed pair of decision problems; we state it as an application-level result about language-model context, not as new structure in the order. The genuinely new pieces are the framing, the surrogate, and the certificate.

## 3 Formalization

§2 placed the contribution against prior work; the next four sections make it precise, beginning here with the objects. We model the choice of *what context to supply* a language model as a comparison of statistical experiments on the task’s latent requirement. This section fixes the objects and, crucially, separates two decision rules that the informal literature silently conflates: the *Bayes-optimal* rule, against which the classical Blackwell order is defined, and the *fixed model* rule, which is the object every empirical quantity in this paper actually measures. Keeping these distinct is what makes the bridge from theorem to experiment rigorous.

### 3.1 Tasks, latent requirements, and context sources

A *task*  $T$  fixes a latent requirement  $S_T$ , the object a deterministic verifier  $v_T$  checks (e.g. the intended program behaviour). We take  $S_T$  to be a *vector of coordinates*

$$S = (S_1, S_2, \dots, S_m) \in \mathcal{S} = \mathcal{S}_1 \times \dots \times \mathcal{S}_m, \quad (1)$$

where each coordinate  $S_j$  encodes one logically independent requirement (in our instantiation,  $S_1$  is the private call-contract convention and  $S_2$  is the private wire-encoding convention; see §7). Modelling  $S$  as a product of coordinates, rather than an opaque atom, is the technical device that makes the experiment model well posed for fixed-string sources; we return to this in §3.2 and Lemma 1.

**Definition 1** (Context source as an experiment). A *context source*  $W$  is a Markov kernel  $\kappa_W : \mathcal{S} \rightarrow \Delta(\mathcal{W})$ , i.e. a statistical experiment indexed by the latent parameter  $S$ , emitting a signal  $w \sim \kappa_W(\cdot \mid S)$  in a signal alphabet  $\mathcal{W}$  (the source’s text). The experiment is identified with its kernel; the sample space is  $\mathcal{W}$  and the observed datum is the supplied text  $w$ .

**Definition 2** (Decision rules and PASS). A *decision rule* is a (possibly randomized) map  $\delta : (T, w) \mapsto \hat{S}$  producing a candidate solution. Utility is the binary verifier  $u(\hat{S}, S) = v_T(\hat{S}) \in \{0, 1\}$ . Two decision rules matter:

- (i) the *model rule*  $\delta_M$  realized by a fixed language model  $M$  at a fixed decoding configuration  $\theta$  (model weights and sampling parameters; see §3.3). Its realized pass rate on task  $T$  under source  $W$  is

$$\text{PASS}_M(W, T) := \mathbb{E}[v_T(\delta_M(T, w))], \quad w \sim \kappa_W(\cdot | S_T); \quad (2)$$

- (ii) the *Bayes rule*  $\delta_T^*$ , the (utility-optimal) decision rule for task  $T$  given a prior over  $S$ , achieving the optimal value  $V^*(W, T) := \sup_{\delta} \mathbb{E}[v_T(\delta(T, w))]$ .

The distinction in Definition 2 is load-bearing. Blackwell’s theorem (§4) is a statement about  $V^*$  (the optimal user of an experiment); every number we report,  $\text{PASS}_M$ , the deficiency estimator of §6, the interference statistic of §6.3, and the entire six-model program of §7, is a statement about the single fixed rule  $\delta_M$ . We therefore develop the order in two registers and prove an explicit bridge between them (Lemma 2).

### 3.2 Sources as deterministic projection experiments

In our instantiation each context source is a *fixed string*: a snippet stating the private call contract ( $W_1$ ), or the private wire-encoding invariant ( $W_2$ ). A naive reading treats  $\kappa_W(\cdot | S)$  as a constant kernel  $\delta_{w_0}$  independent of  $S$ . That reading is degenerate: a constant kernel is Blackwell-equivalent to the null experiment, carries zero information, and induces the trivial  $\sigma$ -algebra, which would make any garbling argument vacuous. We avoid this by modelling each fixed source as a *deterministic experiment that observes a projection of the coordinate vector* (1).

**Definition 3** (Projection sources). A source  $W$  that reveals the coordinates indexed by a set  $A \subseteq \{1, \dots, m\}$  is the deterministic kernel

$$\kappa_W(\cdot | S) = \delta_{\pi_A(S)}, \quad \pi_A(S) := (S_j)_{j \in A}, \quad (3)$$

i.e. supplying  $W$  lets a decision-maker condition on  $\pi_A(S)$  and on nothing outside  $A$ . Write  $\mathcal{F}_W := \sigma(\pi_A)$  for the sub- $\sigma$ -algebra of  $\mathcal{S}$  generated by the projection. In our instantiation  $W_1$  reveals coordinate  $S_1$  (the call contract,  $A_1 = \{1\}$ ),  $W_2$  reveals  $S_2$  (the wire invariant,  $A_2 = \{2\}$ ), and the superset  $W_{1+} = W_2 + W_1$  reveals both ( $A_{1+} = \{1, 2\}$ ).

Under Definition 3 a fixed string is *not* an uninformative constant: it is a deterministic experiment whose informativeness is exactly the resolving power of its  $\sigma$ -algebra  $\mathcal{F}_W$ . The signal alphabet is the range of the projection, and the text “varies with the task” precisely because the realized coordinate values  $\pi_A(S_T)$  differ across task instances  $T$ . This is the well-posed reading that makes the comparison of §4 a comparison of experiments rather than a metaphor, and it is what a rigorous treatment of deterministic sources requires [Torgersen, 1991].

### 3.3 The conditioning parameter $\theta$

The model rule  $\delta_M$  carries an implicit parameter  $\theta$ : the model’s weights and decoding configuration, equivalently its *prior over conventions* acquired in pretraining. We write  $I(S; W | \theta)$  for the model-relative value of a private source, the information  $W$  supplies *beyond what  $\theta$  already encodes*. When  $\theta$  already “knows” a convention (the model can guess it without the source),  $W$ ’s marginal value collapses; when the convention is genuinely unknown to  $\theta$ ,  $W$  is maximally valuable. This

$\theta$ -dependence is not a nuisance: it is the formal home of the *model-relative value of private context* measured in §7, where the guessability of a fixed convention spans 0% to 75% across vendors. Throughout, all realized-PASS quantities are implicitly indexed by  $\theta$ ; we suppress it except where the model-relativity is the point.

## 4 The Blackwell order over context sources

With sources modeled as experiments (§3), the classical comparison of experiments applies to them directly. We state the order, record that it is genuinely partial (the regime our experiments target), and, because those experiments measure a fixed model rather than the idealized decision-maker the classical theorem assumes, build an explicit bridge between its two registers.

### 4.1 The order and its two registers

**Definition 4** (Garbling order).  $W_1 \succeq_B W_2$  ( $W_1$  *Blackwell-dominates*  $W_2$ ) iff there is a Markov kernel (garbling)  $G : \mathcal{W}_1 \rightarrow \Delta(\mathcal{W}_2)$  with  $\kappa_{W_2} = G \circ \kappa_{W_1}$ ; i.e.  $W_2$ 's signal is a noised, post-processed version of  $W_1$ 's, computable from  $W_1$ 's signal alone without further access to  $S$ .

The classical content of the order is the Blackwell–Sherman–Stein theorem, which we state with hypotheses for a self-contained treatment.

**Theorem 1** (Blackwell–Sherman–Stein; universal dominance for the optimal rule). *Let  $W_1, W_2$  be experiments on a common parameter space  $\mathcal{S}$  with standard Borel signal spaces. The following are equivalent:*

- (i)  $W_1 \succeq_B W_2$  (Definition 4);
- (ii) *for every prior  $\mu$  on  $\mathcal{S}$ , every finite decision space, and every bounded loss  $\ell$ , the Bayes risk under  $W_1$  is no larger than under  $W_2$ ; equivalently the optimal value satisfies  $V^*(W_1, T) \geq V^*(W_2, T)$  for every task  $T$  with bounded utility.*

*Randomized decision rules are admitted on both sides; if the dominated experiment is realized with a randomized garbling, no loss of generality is incurred by allowing randomized rules in (ii).*

**Corollary 1** (Universal context dominance, C1). *If  $W_1 \succeq_B W_2$  then, for the optimal user, supplying  $W_1$  is at least as good as supplying  $W_2$  on every task, prior, and bounded utility, a decision-independent, for-all-tasks guarantee that no task-specific scalar score (relevance, mutual information,  $\mathcal{V}$ -usable information) provides.*

*Remark 1* (The order is partial; no lattice).  $\succeq_B$  is a genuine partial order: there exist pairs  $W_1, W_2$  with neither  $W_1 \succeq_B W_2$  nor  $W_2 \succeq_B W_1$ , and the order does not form a lattice [Torgersen, 1991]. Incomparability is exactly the regime our experiments target.

### 4.2 Deterministic sources: incomparability via $\sigma$ -algebra non-refinement

For deterministic projection sources (Definition 3) the garbling order reduces to refinement of the generated  $\sigma$ -algebras, which makes incomparability checkable by a purely structural argument rather than by an empirical divergence.

**Lemma 1** (Refinement characterization for projection sources). *Let  $W_A, W_B$  be projection sources revealing coordinate sets  $A, B$  (Definition 3), with coordinates  $S_1, \dots, S_m$  mutually independent under some full-support prior and each  $S_j$  non-degenerate. Then:*



- (a)  $W_A \succeq_B W_B \iff \mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A} \iff B \subseteq A$ ;
- (b) *consequently  $W_A$  and  $W_B$  are Blackwell-incomparable iff  $A \not\subseteq B$  and  $B \not\subseteq A$ , i.e. each reveals at least one coordinate the other hides.*

Lemma 1 is the formal version of the garbling argument used in §7: from the wire format ( $W_2$ , revealing  $S_2$ ) one cannot manufacture the private arg-order / required-keyword / exception-type facts ( $S_1$ ), and from the call contract ( $W_1$ , revealing  $S_1$ ) one cannot manufacture the cents-scale / check-digit facts ( $S_2$ ). With  $A_1 = \{1\}$ ,  $A_2 = \{2\}$  neither set contains the other, so  $W_1, W_2$  are Blackwell-incomparable by construction (and  $W_{1+}$ , revealing  $\{1, 2\}$ , dominates both). This guarantees the *theoretical* incomparability the experiments then exhibit empirically.

### 4.3 Bridging the optimal-rule order to the fixed model

Theorem 1 concerns  $V^*$ . Our experiments measure  $\text{PASS}_M$  for a single fixed, generally sub-optimal rule  $\delta_M$ . The two are not interchangeable: an inequality between optimal values need not hold for a fixed rule, and, as we show empirically, the fixed rule can even *violate* monotonicity that the optimal value obeys. We therefore introduce a  $\delta_M$ -restricted order and state precisely what it does and does not inherit from Theorem 1.

**Definition 5** ( $\delta_M$ -restricted,  $\mathcal{D}$ -restricted order). Fix the model rule  $\delta_M$  and a finite task set  $\mathcal{D}$ . Say  $W_1 \succeq_{M, \mathcal{D}} W_2$  iff  $\text{PASS}_M(W_1, T) \geq \text{PASS}_M(W_2, T)$  for every  $T \in \mathcal{D}$ . Call  $W_1, W_2$  ( $M, \mathcal{D}$ )-*incomparable* iff  $W_1 \not\succeq_{M, \mathcal{D}} W_2$  and  $W_2 \not\succeq_{M, \mathcal{D}} W_1$ , i.e. each strictly out-performs the other on some task in  $\mathcal{D}$ .

$\succeq_{M, \mathcal{D}}$  is the order our certificate measures. It is decision-restricted (one fixed rule) and distribution-restricted (a fixed finite  $\mathcal{D}$ ); within those restrictions it is still a genuine, *for-all-tasks-in- $\mathcal{D}$*  dominance statement, not a scalar.

**Lemma 2** (One-way bridge under coordinate separation). *Neither order implies the other unconditionally; each direction of the bridge carries an explicit hypothesis.*

- (a) *Let  $W_1, W_2$  be projection sources revealing coordinate sets  $A_1, A_2$  (Definition 3), and let  $T_a, T_b \in \mathcal{D}$  be tasks whose verifier-decisive coordinates lie in  $A_1 \setminus A_2$  and  $A_2 \setminus A_1$  respectively, with guessing caps  $g_a := V^*(W_2, T_a)$  and  $g_b := V^*(W_1, T_b)$ , the optimal values achievable without the decisive coordinate. If  $\text{PASS}_M(W_1, T_a) > g_a$  and  $\text{PASS}_M(W_2, T_b) > g_b$ , then  $W_1, W_2$  are Blackwell-incomparable.*
- (b) *The converse fails in general: Blackwell incomparability concerns  $V^*$ , and a fixed sub-optimal  $\delta_M$  may fail to realize the optimal-value crossover on  $\mathcal{D}$ . It holds under the value-attainment hypothesis (Assumption 1).*

*The separation hypothesis in (a) is necessary, and our own measurements witness the necessity: ( $W_1, W_{1+}$ ) is ( $M, \mathcal{D}$ )-incomparable on the measured cells ( $W_1$  wins the API cells,  $W_{1+}$  the encoding cell) yet Blackwell-comparable by construction ( $W_{1+} \succeq_B W_1$ ), precisely because the deprived-side premise fails there,  $W_{1+}$  contains  $W_1$ 's coordinate, so the API-side cap is not small. A bare  $\text{PASS}_M$  crossover, without the separation hypothesis, witnesses only the sub-optimality of  $\delta_M$  (Theorem 3).*

Intuitively: a fixed-rule crossover proves structural incomparability only when each winning source reveals a coordinate the losing source genuinely lacks; otherwise the crossover may witness nothing more than the rule's own incoherence.

**Assumption 1** (Value attainment on the witnessing tasks). There exist tasks  $T_a, T_b$  and a margin  $\gamma > 0$  such that  $\delta_M$  attains  $\text{PASS}_M(W_1, T_a) - \text{PASS}_M(W_2, T_a) \geq \gamma$  and  $\text{PASS}_M(W_2, T_b) - \text{PASS}_M(W_1, T_b) \geq \gamma$ ; i.e. the fixed model actually exploits each source’s private coordinate on the task that requires it. Under Assumption 1 with  $\{T_a, T_b\} \subseteq \mathcal{D}$ ,  $W_1$  and  $W_2$  are  $(M, \mathcal{D})$ -incomparable.

Lemma 2 is the keystone. In the direction the empirical program needs, it says: a certified  $(M, \mathcal{D})$ -crossover on tasks whose decisive coordinates are *separated* across the two sources, with the deprived side’s optimal value capped below the measured pass rates, implies genuine Blackwell incomparability. Our construction is designed to satisfy the hypothesis (each cell’s decisive private coordinate is revealed by exactly one source, Lemma 1), and its observable signature, deprived-arm pass rates flooring at the guessing cap ( $W_2$  at 0% on the API cells,  $W_1$  at 0% on the encoding cell, **none** near 0), is verified per cell in §7; where a model’s prior partially recovers a convention (the  $\theta$ -dependence of §3.3), the cap premise weakens and we do not certify on those cells. The reverse implication, predicting the realized crossover from the abstract theorem, requires Assumption 1, which our task design enforces and our measurements verify.

## 5 The incomparability theorem

§4 showed the order is genuinely partial; we now establish the consequence that motivates the whole paper: no scalar can rank a Blackwell-incomparable pair. We *split* the claim into a clean theorem for query-*independent* source scores  $\varphi(W)$  and a separate, explicitly empirical statement for query-*conditioned* scores  $\varphi(W, T)$  (the relevance score practitioners actually use); the two must not be conflated.

### 5.1 Query-independent scores: the theorem

**Theorem 2** (No query-independent scalar ranks incomparable sources, C1). *Let  $W_1, W_2$  be Blackwell-incomparable. Let  $\varphi : \{\text{sources}\} \rightarrow \mathbb{R}$  be any query-independent scalar score (a single number per source: relevance to a fixed corpus, mutual information  $I(S; W)$ ,  $\mathcal{V}$ -usable information, perplexity-gap, expected information gain under a fixed prior). Then there exist tasks  $T_a, T_b$  such that the ranking of  $\{W_1, W_2\}$  induced by  $\varphi$  disagrees with the optimal-value ranking on at least one of  $T_a, T_b$ ; and, under Assumption 1, with the realized-PASS<sub>M</sub> ranking on at least one of them.*

*Proof.* By incomparability, neither source garbles the other. By the contrapositive of Theorem 1, “not  $W_1 \succeq_B W_2$ ” yields a task  $T_b$  (a prior plus bounded utility) with  $V^*(W_2, T_b) > V^*(W_1, T_b)$ , and “not  $W_2 \succeq_B W_1$ ” yields a task  $T_a$  with  $V^*(W_1, T_a) > V^*(W_2, T_a)$ ; both inequalities are *strict*, so ties do not arise. A query-independent  $\varphi$  assigns the single ordered pair  $(\varphi(W_1), \varphi(W_2))$ , fixing one total order of  $\{W_1, W_2\}$  *independent of the task*. That fixed order agrees with the optimal-value order on at most one of  $T_a, T_b$  (they point opposite ways), hence is violated on the other. The PASS<sub>M</sub> conclusion follows by replacing  $V^*$  with PASS<sub>M</sub> via Assumption 1, which guarantees the realized rule reproduces both strict inequalities with margin  $\gamma$ .  $\square$

*Remark 2* (Scope). Theorem 2 rules out exactly the class of *source-level* scalars: any  $\varphi$  that is a fixed number per source, regardless of task.  $\mathcal{V}$ -usable information, mutual information, perplexity-gap, and (fixed-prior) EIG all live in this class, so they are corollary victims of the theorem, not competitors (Proposition 1). The theorem is mathematically a one-line consequence of Blackwell’s theorem; the contribution is the *application and framing*, reading cross-task non-transfer of relevance as the signature of a partial order, not new core mathematics [Blackwell, 1953, Torgersen, 1991].

**Proposition 1** ( $\mathcal{V}$ -usable information is Blackwell-monotone). *Let  $\mathcal{V}$ -usable information  $I_{\mathcal{V}}(W \rightarrow S)$  be defined, after Ethayarajh et al. [2022], as the reduction in expected predictive Bayes risk of the best predictor in a fixed family  $\mathcal{V}$  when given the signal  $w$  versus the null signal. If  $W_1 \succeq_B W_2$  and the predictor family  $\mathcal{V}$  is closed under pre-composition with measurable garblings, then  $I_{\mathcal{V}}(W_1 \rightarrow S) \geq I_{\mathcal{V}}(W_2 \rightarrow S)$ . Hence  $I_{\mathcal{V}}$  is a Blackwell-monotone scalar: it respects  $\succeq_B$  but collapses it to one real number.*

*Remark 3* (Scope of Proposition 1). The closure hypothesis (a family closed under pre-composition with garblings) is the natural sufficient condition and holds for predictor families rich enough to apply a fixed measurable post-processing; for narrower families monotonicity may fail and we do not claim it. We prove monotonicity for the closest neighbour,  $\mathcal{V}$ -usable information, and conjecture, rather than prove, the analogue for mutual information, perplexity-gap, and fixed-prior EIG, each of which is a data-processing-monotone functional of the same kernels.

## 5.2 Query-conditioned scores: the relevance trap (empirical)

A query-conditioned score  $\varphi(W, T)$ , e.g. the lexical or embedding relevance of a source to the task prompt, which is what retrieval and RAG actually deploy, induces a *different* order per task and is therefore *not* a single fixed total order. Theorem 2 does *not* cover it. We make two precise statements instead.

**Proposition 2** (Constrained impossibility for feature-measurable scores). *Suppose  $\varphi(W, T)$  is measurable with respect to a fixed feature map of the pair  $(W, T)$  that is topical: it depends on  $W, T$  only through surface lexical overlap and is invariant to the verifier-decisive coordinate of  $S_T$ . If there is a task  $T_c$  whose decisive coordinate is supplied by  $W_1$  but whose surface lexicon is saturated with  $W_2$ ’s vocabulary, then  $\varphi(W_2, T_c) > \varphi(W_1, T_c)$  while the realized utility ranks  $W_1 \succ W_2$  on  $T_c$ . Hence no topical query-conditioned score ranks  $\{W_1, W_2\}$  correctly on all tasks.*

*Proof.* By topical invariance,  $\varphi(\cdot, T_c)$  ignores the verifier-decisive coordinate and orders by lexical overlap, which favours  $W_2$  on  $T_c$  by construction; utility favours  $W_1$  because  $T_c$ ’s PASS requires  $W_1$ ’s coordinate. The two orders disagree on  $T_c$ .  $\square$

Proposition 2 is the strongest *general* statement we can make about query-conditioned scores: it holds for any score in the topical (lexical-feature) class, but not for an arbitrary learned reranker that could, in principle, read the private coordinate. We therefore treat the defeat of query-conditioned relevance as an *empirical* result, demonstrated for the concrete lexical-cosine score practitioners use. In §7 the `trap_store_wire` task realizes  $T_c$ : bag-of-words cosine gives  $\varphi(W_1) = 0.36 < \varphi(W_2) = 0.44$  (ranking  $W_2 \succ W_1$ ), yet realized PASS ranks  $W_1$  at 96%–100% versus  $W_2$  at 0–10% across all six models. The same mis-rank is realized by representative open-weight learned rankers: two dense bi-encoders (BGE, E5) and a cross-encoder reranker all prefer  $W_2$  on the trap (§7.4.2, Table 3), so the topical class is not a lexical strawman: its failure mode is shared by the ranker families retrieval systems actually deploy. We also measure the boundary from the other side: an LLM *listwise* reranker, which reads both documents and can therefore condition on the private coordinate itself, ranks the trap correctly (30/30). This is the behaviour Proposition 2 predicts, its hypothesis is topicality (invariance to the verifier-decisive coordinate), and a ranker that reads the coordinate is simply outside the impossibility class.

## 6 The decision-restricted deficiency surrogate and its uniform certificate

The impossibility of §5 concerns the idealized user of a source; every quantity we report, by contrast, comes from a single fixed model. This section builds the instrument that connects the two: a surrogate that measures the order on the fixed rule, and a distribution-free certificate that makes each comparison auditable on a live model.

### 6.1 The surrogate and its correct name

Exact garblings over text are uncomputable and  $\succeq_B$  is partial, so we measure the  $\delta_M$ -restricted order of Definition 5. Fix the model rule  $\delta_M$  and a finite task set  $\mathcal{D}$  and define the *decision-restricted value-deficiency* (a one-sided worst-case regret of  $W_1$  relative to  $W_2$  on  $\mathcal{D}$ ):

$$\hat{\delta}_{\mathcal{D}}(W_1, W_2) := \max\left(0, \sup_{T \in \mathcal{D}} [\text{PASS}_M(W_2, T) - \text{PASS}_M(W_1, T)]\right), \quad (4)$$

estimated per cell by  $\text{PASS}_M(W, T) \approx k_{W,T}/n$  from  $n$  completions. Then  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$  certifies empirical dominance  $W_1 \succeq_{M, \mathcal{D}} W_2$ , and mutual positivity  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$  certifies  $(M, \mathcal{D})$ -incomparability.

*Remark 4* (This is *not* Le Cam deficiency). The Le Cam deficiency  $\delta(\mathcal{E}, \mathcal{F}) = \inf_G \sup_{\text{loss}} [R_{G \circ \mathcal{E}} - R_{\mathcal{F}}]$  minimizes over garblings  $G$  and takes a supremum over a class of losses [Le Cam, 1964, 1986, Torgersen, 1991]. The quantity in (4) performs no garbling minimization and uses a single fixed rule  $\delta_M$  over a fixed finite task set; it is a one-sided value/regret gap, not a deficiency. No general inequality relates  $\hat{\delta}_{\mathcal{D}}$  to the true Le Cam deficiency for a fixed sub-optimal rule (it can be larger or smaller in either direction). We therefore name (4) the *decision-restricted value-deficiency* and use “Le Cam deficiency” only as the conceptual analogy that motivates a graded relaxation of the partial order. The one direction that *does* transfer is qualitative:  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$  realizes the for-all-tasks-in- $\mathcal{D}$  dominance of Definition 5, the measurable shadow of Corollary 1.

### 6.2 The uniform certificate

The surrogate (4) selects the *argmax-gap* task, so a pointwise confidence statement on a single task ignores the selection. We pay the multiplicity with an exact Clopper–Pearson interval per cell and a union bound, which is distribution-free and finite-sample valid.

**Proposition 3** (Uniform incomparability certificate). *Within each cell  $(W, T)$  draw  $n$  i.i.d. Bernoulli PASS outcomes (see Assumption 2). Let  $[L_{W,T}, U_{W,T}]$  be the exact (two-sided) Clopper–Pearson interval [Clopper and Pearson, 1934] for  $p_{W,T} = \text{PASS}_M(W, T)$  at level  $\alpha = \eta/(2|\mathcal{D}|)$ . By the Bonferroni/union bound [Bonferroni, 1936] over the  $2|\mathcal{D}|$  proportions  $\{p_{W_1,T}, p_{W_2,T}\}_{T \in \mathcal{D}}$ , all  $2|\mathcal{D}|$  intervals hold simultaneously with probability at least  $1 - \eta$ . On that event,*

$$L_{\mathcal{D}}(W_1, W_2) := \max_{T \in \mathcal{D}} [L_{W_2,T} - U_{W_1,T}] \quad (5)$$

*is a  $(1 - \eta)$  lower confidence bound on  $\sup_{T \in \mathcal{D}} [p_{W_2,T} - p_{W_1,T}]$ . Hence  $L_{\mathcal{D}}(W_1, W_2) > 0$  **certifies**  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$  at confidence  $\geq 1 - \eta$ . Both directions  $L_{\mathcal{D}}(W_1, W_2)$  and  $L_{\mathcal{D}}(W_2, W_1)$  read off the same  $2|\mathcal{D}|$  intervals, so the joint incomparability claim  $\{\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0\}$  holds at  $\geq 1 - \eta$  with no additional correction. The matching quantity  $U_{\mathcal{D}}(W_1, W_2) := \max_{T \in \mathcal{D}} [U_{W_2,T} - L_{W_1,T}]$  is, on the same event, a  $(1 - \eta)$  upper confidence bound on the same supremum; the dominance control uses it to certify  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$  up to the equivalence margin.*

A naive single-sample DKW bound under-covers this supremum, because the object is a maximum of differences of binomial means across distinct cells and the certificate selects the argmax-gap cell, so the multiplicity must be paid explicitly rather than absorbed into a single-sample envelope (Appendix A); the resulting construction aligns with recent statistical guidance for black-box LLM evaluation (Appendix A).

**Assumption 2** (Sampling regime). The  $n$  outcomes within a cell are i.i.d. Bernoulli draws of  $v_T(\delta_M(T, w))$ . This holds when completions are sampled independently at fixed temperature/top- $p$  with independent randomness per draw; for the single-shot arms (one completion + deterministic code extraction per draw) the draws are manifestly independent. For the agentic arms (`claude-p`, multi-turn) independence is across *trajectories*: each of the  $n$  draws is a fresh trajectory with no shared state, so the binomial model applies to the trajectory-level PASS outcome; we report decoding configuration and defend this independence in §7. Where draws are near-deterministic ( $p \in \{0, 1\}$  stark cells), Clopper–Pearson collapses and the certificate is conservative.

### 6.3 The interference certificate (anti-monotonicity)

The deficiency surrogate also detects a phenomenon the optimal-value order forbids but a fixed rule can exhibit: supplying a *superset* source  $W_{1+}$  (revealing strictly more coordinates than  $W_1$ ; Definition 3) can *degrade* performance. Define the per-task interference

$$\Psi_T := \text{PASS}_M(W_{1+}, T) - \text{PASS}_M(W_1, T) - \text{PASS}_M(W_2, T) + \text{PASS}_M(\text{none}, T), \quad (6)$$

the second-difference (interaction) of adding  $W_2$  on top of  $W_1$ . We write the interaction as  $\Psi_T$  (rather than a symbol  $I$  that would collide with mutual information  $I(S; W \mid \theta)$ ). A certified  $\Psi_T < 0$  on a cell whose  $W_1$  arm is at ceiling is the localized statement that “more strictly-more-informative context hurts.”

**Proposition 4** (Interference certificate). *Fix a target task  $T^* \in \mathcal{D}$  chosen before unblinding. With per-cell exact Clopper–Pearson endpoints  $\text{CP}_{\text{lo}}, \text{CP}_{\text{hi}}$  at level  $\alpha = \eta/4$  on the four arms  $\{W_{1+}, W_1, W_2, \text{none}\}$  of  $T^*$ ,*

$$\bar{\Psi}_{T^*} := \text{CP}_{\text{hi}}(p_{W_{1+}}) - \text{CP}_{\text{lo}}(p_{W_1}) - \text{CP}_{\text{lo}}(p_{W_2}) + \text{CP}_{\text{hi}}(p_{\text{none}}) \quad (7)$$

*is a  $(1 - \eta)$  upper confidence bound on  $\Psi_{T^*}$ . Hence  $\bar{\Psi}_{T^*} \leq -\tau$  certifies harmful non-monotonicity  $\Psi_{T^*} \leq -\tau$  at confidence  $\geq 1 - \eta$ . If instead the harmful cell is selected from  $|\mathcal{D}|$  candidates after unblinding, use  $\alpha = \eta/(4|\mathcal{D}|)$ ; this is the same selection correction the certificate applies to the deficiency surrogate (Remark 6), and omitting it would repeat the multiplicity error we criticize.*

**Remark 5** (The leaky-baseline limitation, stated as a protocol). The upper bound (7) includes  $+\text{CP}_{\text{hi}}(p_{\text{none}})$ , so it can certify harm only on cells whose **none**-baseline floors near 0 (the convention is unguessable for the model under test). When the baseline is non-zero, e.g. `api_argorder` at **none** = 75% for GPT-5.5, 45% for Kimi, 15% for Opus, the bound is mechanically slack and harm is not certifiable even when the point estimate is strongly negative. We therefore adopt the explicit protocol: *certify  $\Psi_T$  only on cells whose **none**-baseline floors near 0 for the model under test*, and report, per model, which cell carried the certificate and what its baseline was. This makes the cell selection a stated rule rather than an outcome-driven choice. The same model-relative guessability that defines the value of private context (§3.3) thus governs which cells can carry the interference certificate.

## 6.4 Anti-monotonicity as information non-monotonicity of $\delta_M$

A certified  $\Psi_{T^*} < 0$  on a ceiling- $W_1$  cell is more than a curiosity: it is a falsification of Blackwell superset-dominance *for the fixed model rule*, and it pinpoints exactly where the optimal-rule order (Theorem 1) and the realized rule diverge.

**Theorem 3** (Information non-monotonicity of the model rule). *Let  $W_{1+}$  reveal a superset of  $W_1$ ’s coordinates, so  $W_{1+} \succeq_B W_1$  (Lemma 1). Any Blackwell-coherent rule, one whose value is monotone under  $\succeq_B$ , as the Bayes rule  $\delta^*$  is by Theorem 1, satisfies  $\text{value}(W_{1+}, T) \geq \text{value}(W_1, T)$  for every task  $T$ . If a fixed model rule  $\delta_M$  has a task  $T^*$  with  $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$ , then  $\delta_M$  is not Blackwell-coherent: it violates monotonicity under the garbling order on  $T^*$ . In particular no monotone-submodular scalar context-value score can predict  $\delta_M$ ’s behaviour on  $T^*$ .*

Theorem 3 converts the empirical collapse into a structural statement: the harm is not noise but a witness that real LLM decision rules are not Blackwell-coherent, which is precisely the gap Lemma 2(b) warned cannot be closed without value attainment. The deficiency surrogate detects this reversal because it tracks realized PASS, whereas every monotone scalar must miss it by construction.

## 6.5 Family-wise error across the certified claims

Each certificate above controls error *marginally* at  $\eta$  for its own claim (incomparability via Proposition 3; interference via Proposition 4). When several claims are read off the *same*  $n$ -sample run at the same  $\eta$ , incomparability (both directions, on one shared event  $E$ ), one-directional dominance, and the interference cells, the conjunction over distinct claims is *not* jointly controlled at  $\eta$  unless the budget is split across claims. We state results as marginally certified per claim and, where a joint statement is wanted, allocate  $\eta$  across the  $C$  certified claims (e.g.  $\eta/C$  each) so the global family-wise error is  $\leq \eta$ . The incomparability pair is the one exception that is genuinely joint at  $\eta$  on a single event, as proved in Proposition 3; all other cross-claim conjunctions on a shared run are reported as marginal.

The two certificates have genuinely different effective- $n$  requirements, which is why the consolidated record marks Opus incomparability as certified and Opus interference as reproduced but not certified; the power analysis is in Appendix A.

# 7 Experiments

With the order (§4), its impossibility consequence (§5), and the certificate that measures it (§6) in hand, we now instantiate the theory on real language models. The design has three commitments. First, every reported number is an empirical PASS rate from live model calls verified by an executable test harness; no value is modeled, simulated, or extrapolated. Second, the two context sources are constructed to be *Blackwell-incomparable by construction* (§7.1), so the incomparability theorem (Theorem 2) has a non-vacuous instance to bite on. Third, every certified claim is reported with the exact sample size, the Clopper–Pearson endpoints, and the per-cell PASS counts from which it is derived (§7.4), so the certificate is auditable rather than asserted.

## 7.1 Design: two incomparable context sources

We define two *private*, *unguessable* conventions on the *same* module `ledger`, matched on every nuisance axis (length, roughly three sentences, exactly one private convention each, same module,

same task shape). Because both are on-topic for any “ledger” task, any lexical, embedding, mutual-information, or  $\mathcal{V}$ -usable score treats them as equally relevant candidates, the precondition for Theorem 2 to apply.

$W_1$ , **API call contract**. `ledger.post(account, amount, *, memo)`: `account` first, `amount` second positional, `memo` a *required keyword-only* argument; returns an integer entry `id`  $\geq 1$ ; raises `ledger.PostError` (not `ValueError/KeyError`) on failure.

$W_2$ , **wire-encoding invariant**. Amounts serialize to `<sign><minor-unit-digits>#<check>`, where `check` = (sum of digits) mod 10; e.g.  $\$12.30 \mapsto "+1230\#6"$  and  $-\$0.07 \mapsto "-7\#7"$ . Decoding rejects a wrong check digit.

$W_{1+}$ , **literal superset**.  $W_{1+} = W_2 ++ W_1$ , the concatenation of  $W_2$ ’s signal with  $W_1$ ’s. As a string superset of  $W_2$ , it Blackwell-dominates  $W_2$  ( $W_{1+} \succeq_B W_2$ ); it is the anti-monotonicity / dominance probe. We also use  $W_{1+}^{\text{rev}} = W_1 ++ W_2$  ( $W_1$  first) as an order control.

**Why incomparable (garbling argument)**. A garbling acts on the *signal alone*. From  $W_2$ ’s wire format one cannot manufacture the private argument order, the required keyword-only `memo`, or the exception type; from  $W_1$ ’s contract one cannot manufacture the cents scale, the `#` separator, or the mod-10 check. The two facts are logically independent coordinates of the latent requirement  $S$ , so neither signal’s  $\sigma$ -algebra refines the other’s, neither sufficiency direction holds, and  $W_1, W_2$  are Blackwell-incomparable. This is the concrete instance of the coordinate/projection construction of §3:  $W_1$  and  $W_2$  are deterministic experiments observing distinct, mutually non-refining projections of  $S$  (Lemma 1).

**Confound defense: the double crossover**. The experiment’s signature is a *double crossover* measured against a **none** baseline (no convention supplied):  $W_1$  lifts the API tasks but *not* the encoding task, while  $W_2$  lifts the encoding task but *not* the API tasks. If the encoding family were merely “harder,”  $W_1$  would not lift it and  $W_2$  would still have to lift its own family from the same floor. Matched **none** baselines that floor near zero on both families therefore establish equal intrinsic hardness, so any divergence is attributable to the matching private fact, not to a difficulty asymmetry.

## 7.2 Tasks

The full design comprises seven tasks with stub-`ledger` verifiers (return code 0 = PASS): four  $W_1$ -decisive API tasks (`api_post_ok`, `api_argorder`, `api_error_type`, `api_memo_required`), two  $W_2$ -decisive encoding tasks (`enc_amount`, `dec_amount`), and one relevance trap (`trap_store_wire`). The verifier stubs a fake `ledger` that accepts *only* the exact private contract (rejecting wrong positional order, a missing `memo=`, or the wrong exception type).

**Certified battery**. The certified runs use a fixed four-cell battery: `api_post_ok`, `api_argorder` (hardened so `memo` arrives through a parameter, forcing reliance on the keyword-only fact), `enc_amount`, and the trap `trap_store_wire`. We disclose this reduction explicitly: the three remaining designed cells (`api_error_type`, `api_memo_required`, `dec_amount`) are not part of the certified battery. The four-cell selection was made to retain at least one  $W_1$ -decisive, one  $W_2$ -decisive, and the trap cell while keeping the cross-model spend tractable; we report below that this is best understood as a fixed adversarial four-cell battery, not a sampled task distribution, and we state the effective support of the deficiency surrogate accordingly (§7.4.1).

**The relevance trap.** The trap `trap_store_wire` is the confound-proof form of Theorem 2. Its prompt is lexically saturated with encoding vocabulary, so a *query-conditioned* lexical relevance score ranks  $W_2 > W_1$ ; but its graded requirement is the  $W_1$  call contract (it must post an already-encoded wire string through `post`), so realized PASS ranks  $W_1 > W_2$ . This defeats the query-conditioned relevance score practitioners actually use, not merely a query-independent ranking. We measure relevance as bag-of-words cosine  $\varphi_{\text{lex}}(W, T)$  between a source and the task prompt, computed *before* the model is run. We use lexical cosine as the *minimal* instantiation of Proposition 2, the cheapest member of the topical class; stronger learned rankers (dense bi-encoders, a cross-encoder, and an LLM listwise reranker) are tested against the same trap separately in §7.4.2, while the theorem (Theorem 2) is the general claim for query-independent scalars.

### 7.3 Methodology

**Models and vendors.** We evaluate six models across four vendors: Anthropic Haiku-4.5, Sonnet-4.6, and Opus-4.8; OpenAI GPT-5.5; DeepSeek-V4-Pro; and Moonshot Kimi-K2.6.

**Harness and arms.** The harness (`tokenbench/measure_blackwell.py`; released with all run logs at <https://github.com/abilous-ti/blackwell-llm-context>) is Python-standard-library only (zero external dependencies); it computes  $\hat{\delta}_{\mathcal{D}}$  in both directions, the uniform Clopper–Pearson/Bonferroni certificate, the interference certificate, and the lexical relevance score, and it implements the launchers for the Anthropic CLI, the Azure Responses API, and OpenAI-compatible chat completions. Each cell is run under the arms  $\{\text{none}, W_1, W_2, W_{1+}\}$ , with  $W_{1+}^{\text{rev}}$  added for the order-control run. PASS is measured over  $n$  completions per cell.

**Agentic vs. single-shot harness (a stated confound).** The Anthropic models run *agentially* (`claude -p`, up to six turns with intermediate verifier feedback); the non-Anthropic models run as *single-shot* completion plus code extraction through an OpenAI-style API. Vendor is therefore confounded with harness in this study: every Anthropic model is agentic and every non-Anthropic model is single-shot. We treat this as a genuine limitation of the cross-vendor *magnitude* comparisons (§8.5); the per-model *binary* claims (incomparability certified, relevance mis-rank, anti-monotonicity certified) are within-model and unaffected. We note that the single-shot setting is arguably the cleaner probe of whether a model uses supplied context, since it removes the agentic-exploration pathway.

**Certificate parameters.** All certificates use confidence budget  $\eta = 0.10$ . Incomparability uses exact Clopper–Pearson intervals at per-cell level  $\alpha = \eta/(2|\mathcal{D}|)$  over the  $2|\mathcal{D}|$  proportions  $\{p_{W_1, T}, p_{W_2, T}\}_{T \in \mathcal{D}}$ , so by the union bound all hold jointly with probability  $\geq 1 - \eta$  and both directions read off the same intervals. The interference certificate uses  $\alpha = \eta/4$  and the threshold  $\tau = 0.30$ , certifying  $\Psi_T \leq -\tau$  via the upper bound of (7). We report each certificate as marginally controlled at  $\eta$ ; we do not claim joint family-wise control across the incomparability, dominance, and interference claims drawn from a single run (§6.5, §8.5). The intervals are degenerate at  $p = 0$  and  $p = 1$ , which is why the stark 100%/0% cells certify incomparability at small  $n$  while the interference contrast, which mixes intermediate baselines, does not.

**Sampling regime.** The certificate models each cell as  $n$  Bernoulli PASS draws (Assumption 2); the four arms within a cell are separate completions, hence independent. We report the decoding parameters, seeding policy, retry policy, and turn budget in Appendix C. The independence



Table 1: Consolidated 6-model / 4-vendor verdict. Anthropic rows are agentic (**claude -p**, up to six turns); the others are single-shot (§7.3). “Anti-mono. certified” counts cells on which the interference upper bound clears  $-\tau = -0.30$  at  $\eta = 0.10$ . “Argorder **none**” is the no-context baseline PASS on **api\_argorder**, the model-relative-value spectrum. All figures are real measured runs; no modeled numbers.

| Claim                             | Haiku     | Sonnet    | Opus       | GPT-5.5     | DeepSeek    | Kimi-K2.6   |
|-----------------------------------|-----------|-----------|------------|-------------|-------------|-------------|
| Vendor                            | Anthropic | Anthropic | Anthropic  | OpenAI      | DeepSeek    | Moonshot    |
| Harness                           | agentic   | agentic   | agentic    | single-shot | single-shot | single-shot |
| $n$ per cell                      | 80        | 40        | 20         | 40          | 40          | 40          |
| Incomp. certified ( $\geq 90\%$ ) | ✓         | ✓         | ✓          | ✓           | ✓           | ✓           |
| $L_{\mathcal{D}}(W_1, W_2)$       | +0.88     | +0.76     | +0.55      | +0.76       | +0.347      | +0.675      |
| $L_{\mathcal{D}}(W_2, W_1)$       | +0.83     | +0.76     | +0.55      | +0.76       | +0.715      | +0.762      |
| Relevance mis-rank (trap)         | ✓         | ✓         | ✓          | ✓           | ✓           | ✓           |
| Anti-mono. certified (# cells)    | 2         | 2         | 0 (repro.) | 1           | 1           | 1           |
| <b>api_argorder none</b>          | ~0%       | 0%        | 15%        | 75%         | 2%          | 45%         |

assumption is most delicate for the agentic arm, where each draw is a multi-turn trajectory; each run is a fresh trajectory with no shared state, and the binomial model underlying every  $L_{\mathcal{D}}$ ,  $U_{\mathcal{D}}$ , and  $\Psi_T$  bound applies at the trajectory level (§8.5).

**Spend.** The record reflects roughly \$210 of Anthropic spend (the live **claude -p** runs; itemized in Appendix C) plus Azure quota for the non-Anthropic models; the Azure API returns no cost field, so the \$210 figure should not be read as the total cost across all six models.

## 7.4 Results

We report four findings: incomparability (§7.4.1), the relevance mis-rank (§7.4.2), anti-monotonicity (§7.4.3), and the model-relative value of private context (§7.4.4). Table 1 consolidates the per-model verdicts; Tables 2, 4, and 5 give the underlying quantities.

### 7.4.1 Incomparability certified on all six models

On every model the empirical deficiency is mutually positive,  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$  and  $\hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$ , and the uniform lower bounds  $L_{\mathcal{D}}$  in both directions clear zero, certifying empirical incomparability at  $\geq 90\%$  (Table 2). The crossover is stark:  $W_1$  solves the API and trap cells and fails encoding, while  $W_2$  solves encoding and fails the API and trap cells. For Haiku at  $n = 80$  the carrying cells are **api\_post\_ok** (**none** 0% /  $W_1$  98% /  $W_2$  0%) and **enc\_amount** (**none** 0% /  $W_1$  0% /  $W_2$  100%); the certificate strengthened from a marginal  $L_{\mathcal{D}} = +0.138$  at  $n = 10$  to  $L_{\mathcal{D}} = +0.877 / +0.831$  at  $n = 80$ , far clear of zero.

Because incomparability is a two-directional claim, each row of Table 2 requires both bounds positive; the smaller of the two is the binding one. The two carrying cells are one API cell and one encoding cell, so the effective support of the surrogate is two cells. We therefore describe  $\mathcal{D}$  as a fixed adversarial four-cell battery rather than a sampled task distribution: the certificate is valid for what it covers, and what it covers is a single hand-built API-versus-encoding opposition replicated across six models.

Table 2: Certified incomparability: uniform  $(1 - \eta)$  lower bounds  $L_{\mathcal{D}}$  in both directions ( $\eta = 0.10$ ).  $L_{\mathcal{D}}(W_1, W_2) > 0$  and  $L_{\mathcal{D}}(W_2, W_1) > 0$  together certify empirical incomparability. Both bounds are read off the same  $2|\mathcal{D}|$  Clopper–Pearson intervals (Proposition 3).

| Model           | Vendor    | $n$ | $L_{\mathcal{D}}(W_1, W_2)$ | $L_{\mathcal{D}}(W_2, W_1)$ |
|-----------------|-----------|-----|-----------------------------|-----------------------------|
| Haiku-4.5       | Anthropic | 80  | +0.877                      | +0.831                      |
| Sonnet-4.6      | Anthropic | 40  | +0.762                      | +0.762                      |
| Opus-4.8        | Anthropic | 20  | +0.552                      | +0.552                      |
| GPT-5.5         | OpenAI    | 40  | +0.762                      | +0.762                      |
| DeepSeek-V4-Pro | DeepSeek  | 40  | +0.347                      | +0.715                      |
| Kimi-K2.6       | Moonshot  | 40  | +0.675                      | +0.762                      |

**Why  $n = 20$  certifies incomparability for Opus but not interference.** Opus is run at  $n = 20$ , which certifies incomparability ( $L_{\mathcal{D}} = +0.552$  both directions) but not anti-monotonicity. This asymmetry is principled, not an inconsistent threshold (Appendix A). The incomparability bound is carried by stark cells whose proportions sit at 0 or 1, where the Clopper–Pearson interval is degenerate and a small  $n$  suffices. The interference bound, by contrast, mixes the  $W_{1+}$ ,  $W_1$ ,  $W_2$ , and **none** arms, including the intermediate Opus baselines (`api_argorder` and `trap none = 15%`), so the four-term Clopper–Pearson sum stays wide at  $n = 20$  (upper bounds  $-0.08/ -0.11$ , not below  $-\tau$ ). Raising Opus to  $n = 80$  is the one remaining gap in the consolidated record.

#### 7.4.2 The relevance score mis-ranks the sources

On the trap, query-conditioned lexical relevance ranks  $\varphi_{\text{lex}}(W_1) = 0.36 < \varphi_{\text{lex}}(W_2) = 0.44$ , so it prefers  $W_2$ ; yet realized PASS ranks  $W_1 \gg W_2$  on all six models (e.g. Haiku 96% vs. 0% at  $n = 80$ ; Sonnet, GPT-5.5, and Kimi 100% vs. 0%; DeepSeek 98% vs. 0%; Opus 100% vs. 10%). On the honest cells relevance *agrees* with PASS, on the API cells relevance ranks  $W_1 > W_2$  and so does PASS, and on the encoding cell relevance ranks  $W_2 > W_1$  and so does PASS, so the score fails *only* where utility diverges from topicality, exactly the regime Theorem 2 and Proposition 2 isolate. The **none** floor on the trap (0% on five models, 15% for Opus) confirms genuine unguessability, and  $W_2$  at 0% (10% for Opus) confirms that encoding knowledge is useless there.

**The ranker spectrum: dense retrievers and a cross-encoder fall for the trap; a listwise reranker escapes it.** To test whether the mis-rank extends beyond bag-of-words, we scored  $\{W_1, W_2\}$  against the probe tasks with representative open-weight rankers, two dense bi-encoders (`bge-small-en-v1.5`, `e5-small-v2`) and a cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`), and with a RankGPT-style LLM *listwise* reranker (Haiku single-shot, seeing the task and *both* sources; both presentation orders;  $n = 30$  judgments per task). Table 3 reports the induced rankings. **Every feature-based ranker mis-ranks the trap:** both dense bi-encoders prefer  $W_2$  on the trap while ranking both honest cells correctly, textbook topical behaviour, and the cross-encoder prefers  $W_2$  on the trap with maximal confidence (logits  $-5.63$  for  $W_1$  vs  $+3.24$  for  $W_2$ ) while being near-tied (and narrowly wrong) on the honest encoding cell. The LLM listwise reranker, which can read the verifier-decisive coordinate rather than match surface features, tracks realized PASS everywhere (30/30 per task, both orders). This is exactly the boundary Proposition 2 draws: the impossibility hypothesis is *topicality*, invariance to the decisive coordinate, and the trap defeats the lexical, dense, and cross-encoder families that satisfy it while sparing the one ranker that reads content. The practical corollary is that escaping the topical class costs a model call per candidate, a probe of the

Table 3: The ranker spectrum on the probe tasks. Each cell is the ranker’s top pick between  $\{W_1, W_2\}$ ; “PASS” is the realized-utility ranking from the certified record. Dense and cross-encoder scores are deterministic; the LLM listwise row pools  $n = 30$  judgments per task (unanimous, both presentation orders).  $\times$  marks a mis-rank. The cross-encoder’s `enc_amount` pick is a near-tie ( $-0.279$  vs  $-0.305$ ).

| Ranker                    | Family           | trap (PASS: $W_1$ ) | api_post_ok (PASS: $W_1$ ) | enc_amount (PA |
|---------------------------|------------------|---------------------|----------------------------|----------------|
| bag-of-words cosine       | lexical          | $W_2 \times$        | $W_1$                      | $W_2$          |
| bge-small-en-v1.5         | dense bi-encoder | $W_2 \times$        | $W_1$                      | $W_2$          |
| e5-small-v2               | dense bi-encoder | $W_2 \times$        | $W_1$                      | $W_2$          |
| ms-marco-MiniLM-L-6-v2    | cross-encoder    | $W_2 \times$        | $W_1$                      | $W_1 \times$   |
| Haiku listwise ( $n=30$ ) | LLM listwise     | $W_1$               | $W_1$                      | $W_2$          |

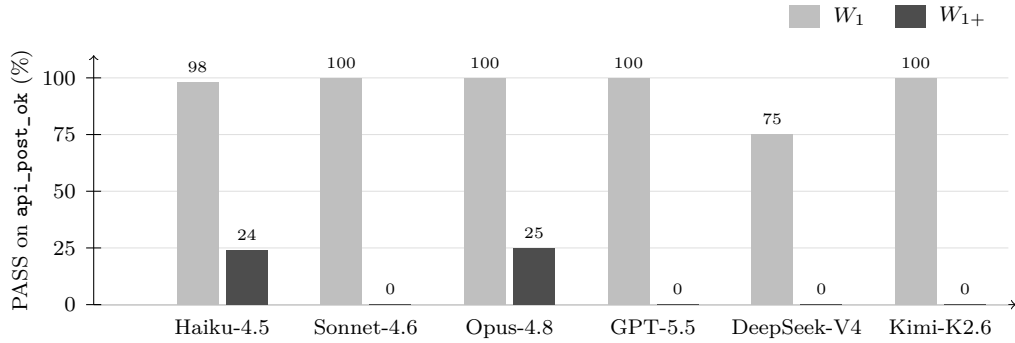


Figure 1: Anti-monotonicity across the capability ladder: PASS on the `api_post_ok` cell under the sufficient source  $W_1$  (light) versus the strict superset  $W_{1+} = W_2 ++ W_1$  (dark), per model (Table 5). The superset *contains*  $W_1$ ’s signal yet collapses performance on every model; the effect is large everywhere but not monotone in capability (Sonnet collapses hardest, Opus least). Anthropic rows agentic, others single-shot; Opus is  $n=20$  (reproduced, not certified).

same kind the deficiency certificate is built from, not a free scalar.

#### 7.4.3 Anti-monotonicity: a certified superset that hurts

The superset  $W_{1+}$ , which literally *contains*  $W_1$ ’s signal, sharply degrades the API cells relative to  $W_1$  alone. The per-task interference  $\Psi_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T) - \text{PASS}(W_2, T) + \text{PASS}(\text{none}, T)$  is certified negative ( $\Psi_T \leq -\tau$ ,  $\tau = 0.30$ ) on at least one cell for five of the six models (Table 4). Opus reproduces the effect with strong point estimates ( $\Psi_T = -0.75 / -0.80$ ) but does not certify at  $n = 20$ . This is a certified empirical falsification of Blackwell superset-dominance on live models (Theorem 3): strictly-more-informative context can hurt, and the deficiency framework detects it precisely where every monotone scalar must miss it. Figure 1 shows the collapse across the six models.

**The empirical order inverts the idealized one.** The idealized Blackwell order on  $\mathcal{C} = \{W_1, W_2, W_{1+}\}$  collapses the choice to the superset:  $W_{1+}$  reveals both coordinates, so  $W_{1+} \succeq_B W_1$  and  $W_{1+} \succeq_B W_2$ , and “more signal is better” keeps  $\{W_{1+}\}$  alone. The certified empirical relation for the fixed rule is the opposite shape:  $W_1$  beats  $W_{1+}$  on the API cells,  $W_{1+}$  beats  $W_1$  on the

*Idealized Blackwell order (optimal user)*    *Certified empirical order (fixed rule  $\delta_M$ , Haiku  $n=80$ )*

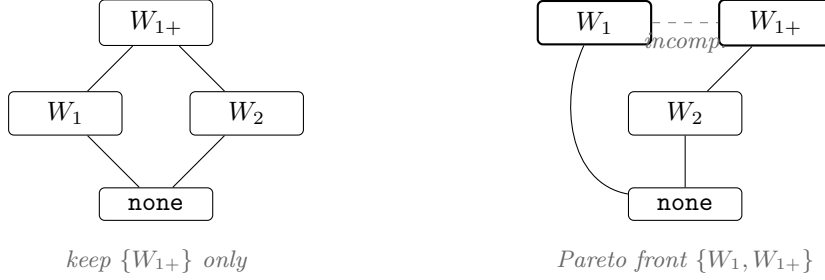


Figure 2: The paper’s thesis in one picture, on the measured  $\mathcal{C} = \{W_1, W_2, W_{1+}\}$  with  $W_{1+} = W_2 ++ W_1$ . **Left:** the idealized Blackwell order for the optimal user;  $W_{1+}$  reveals both coordinates, so “more signal is better” collapses the choice to the superset. **Right:** the certified empirical order for the fixed model rule (Haiku,  $n=80$ ):  $W_1$  beats  $W_{1+}$  on the API cells (98% vs. 24%, anti-monotonicity, Table 4) while  $W_{1+}$  beats  $W_1$  on the encoding cell, so the two are empirically incomparable (dashed) and the correct output is the two-element Pareto front, which no scalar top-1 rule can represent.

encoding cell,  $W_{1+}$  dominates  $W_2$  (while  $W_1$  and  $W_2$  are the certified incomparable pair), so the correct empirical summary is the two-element Pareto front  $\{W_1, W_{1+}\}$ , an object no scalar top-1 rule can represent (Figure 2). This reads directly off the certified  $n=80$  Haiku cells; the selection rule it implies, keep the certified non-dominated set rather than force the superset, is developed in a companion paper.

**The leaky-baseline limitation is a stated protocol.** The interference upper bound includes the term  $+CP_{hi}(p_{\text{none}})$ , so it can certify harm only on cells whose no-context baseline floors near zero. This is exactly why anti-monotonicity certifies on `api_argorder` (hardened) for Haiku (`none` dropped from 40% to 1% after hardening,  $W_1$  still 100%) but not on `api_argorder` for GPT-5.5 (`none` = 75%), Kimi (45%), or Opus (15%). We state this as a protocol up front (Remark 5): we certify anti-monotonicity only on cells whose `none` baseline floors near zero, because a leaky baseline mechanically shrinks  $\Psi_T$ ’s certifiable magnitude. Table 4 therefore reports, per model, which cell carried the certificate and the collapse on it. We also report the  $W_1$  ceiling alongside every  $\Psi_T$ : DeepSeek’s  $W_1$  arm reaches only 75% on `api_post_ok` (an honest capability difference, vs.  $\sim 100\%$  for Claude and GPT), so its  $\Psi_T = -0.75$  is computed from a lower starting point and is not directly magnitude-comparable to Sonnet’s  $100\% \rightarrow 0\%$ .

**Order control.** Reordering the superset does not restore performance. At  $n = 80$  on Haiku, `api_post_ok` gives  $W_{1+}$  24% and  $W_{1+}^{\text{rev}}$  1% (vs.  $W_1$  98%), and `api_argorder` gives  $W_{1+}$  1% and  $W_{1+}^{\text{rev}}$  44% (vs.  $W_1$  100%). On one cell the reversed order is worse and on the other it is better, so order does have a directional effect; but *neither* ordering recovers  $W_1$  alone. The defensible reading is that reordering does not cure the harm, hence the collapse is not a lost-in-the-middle / recency artifact, not that order is irrelevant.

**Runtime fingerprint (a descriptive correlate).** The harmful  $W_{1+}$  arm consistently emits more output tokens than  $W_1$ : between +87% and +252% across the harmful Haiku cells (and, in the full-battery run, +12% even on the still-passing encoding cell), Sonnet +120% / +131%, Opus

Table 4: Anti-monotonicity (interference) certificate on the carrying cell per model.  $\Psi_T = \text{PASS}(W_{1+}) - \text{PASS}(W_1) - \text{PASS}(W_2) + \text{PASS}(\text{none})$ ; “Upper” is the  $\eta/4$  Clopper–Pearson upper bound  $\bar{\Psi}_T$  of (7); the cell is certified harmful when  $\text{Upper} \leq -0.30$ . The  $W_1 \rightarrow W_{1+}$  column reports the PASS collapse. The certificate clears only on cells whose **none** baseline floors near zero (the leaky-baseline limitation; Remark 5).

| Model           | Carrying cell                        | $W_1 \rightarrow W_{1+}$ | $\Psi_T$ | Upper | Certified       |
|-----------------|--------------------------------------|--------------------------|----------|-------|-----------------|
| Haiku-4.5       | <code>api_post_ok</code>             | 98% $\rightarrow$ 24%    | −0.74    | −0.49 | ✓               |
| Haiku-4.5       | <code>api_argorder</code> (hardened) | 100% $\rightarrow$ 1%    | −0.97    | −0.79 | ✓               |
| Sonnet-4.6      | <code>api_post_ok</code>             | 100% $\rightarrow$ 0%    | −1.00    | −0.69 | ✓               |
| Sonnet-4.6      | <code>api_argorder</code>            | 100% $\rightarrow$ 0%    | −1.00    | −0.69 | ✓               |
| GPT-5.5         | <code>api_post_ok</code>             | 100% $\rightarrow$ 0%    | −0.95    | −0.60 | ✓               |
| DeepSeek-V4-Pro | <code>api_post_ok</code>             | 75% $\rightarrow$ 0%     | −0.75    | −0.36 | ✓               |
| Kimi-K2.6       | <code>api_post_ok</code>             | 100% $\rightarrow$ 0%    | −1.00    | −0.69 | ✓               |
| Opus-4.8        | <code>api_post_ok</code>             | 100% $\rightarrow$ 25%   | −0.75    | −0.08 | no ( $n = 20$ ) |
| Opus-4.8        | <code>api_argorder</code>            | 100% $\rightarrow$ 5%    | −0.80    | −0.11 | no ( $n = 20$ ) |

+204%/ + 204%, GPT-5.5 +281%/ + 274%, DeepSeek +93%/ + 166%, Kimi +203%/ + 249%. We report this as a consistent descriptive correlate, not a mechanism: more output on the failing arm could be effect rather than cause, and the +12% inflation on a cell that still passes undercuts a strict “verbose thrashing causes the collapse” reading.

**Robust, but not monotone in capability.** The harm is large and robust across the Anthropic capability ladder but is *not* ordered by capability. On the two API cells the  $W_1 \rightarrow W_{1+}$  collapse is Haiku 98%/100%  $\rightarrow$  24%/1%, Sonnet 100%  $\rightarrow$  0%/0%, and Opus 100%  $\rightarrow$  25%/5%: the middle model (Sonnet) shows the starkest collapse and the strongest model (Opus) less so. An earlier two-point reading suggested the harm “intensifies with strength”; the third point falsifies that, and we explicitly do not claim a capability trend, only robustness across the ladder.

#### 7.4.4 The model-relative value of private context

The “private” conventions are genuinely unguessable for some vendors and partially known to others, and this varies continuously. The **none**-arm baseline on `api_argorder` forms a clean spectrum: Haiku and Sonnet  $\sim 0\%$ , DeepSeek 2%, Opus 15%, Kimi 45%, GPT-5.5 75% (Table 1; Figure 3). Thus the value of the private context, the model-relative quantity  $I(S; W \mid \theta)$  of §3.3, varies continuously across models. It tracks a vendor’s training rather than raw capability: DeepSeek, a different vendor from Anthropic, nonetheless has a tight prior (baselines 0/2/0/0% across `api_post_ok`/`api_argorder`/`enc_amount`/`trap`) like Claude, while GPT-5.5’s prior already holds much of the convention (`api_argorder` 75%, `enc_amount` 55%). Sonnet’s baselines all floor at 0%, so even the stronger Anthropic model does not guess the convention; the value of private context survives the capability ladder. The certified per-model claims rest on cells unguessable *for the model under test* (e.g. GPT-5.5 `api_post_ok none` = 5%, `trap` = 0%).

We note that this cross-vendor magnitude comparison is the part most exposed to the agentic-vs-single-shot confound (§8.5): vendor and harness are not separated in the present runs.

**Auditable per-cell record.** The full per-arm PASS record for every carrying cell is Table 5 in Appendix B, so each certificate can be reconstructed from the reported counts.

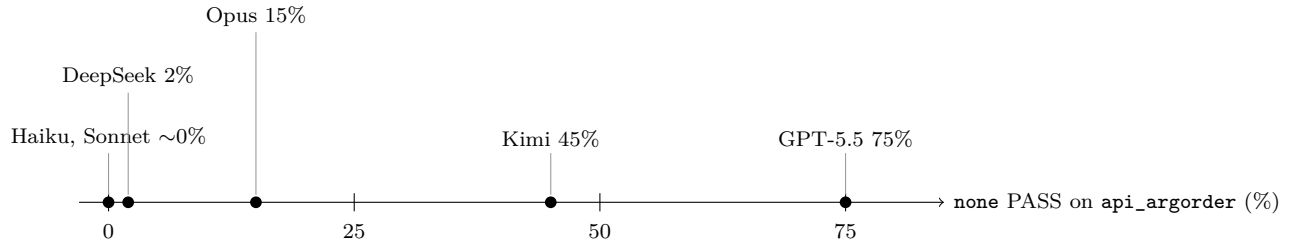


Figure 3: The model-relative value of private context as a spectrum: how often each model reproduces the private convention with *no* context supplied (**none** arm, **api\_argorder**). The value of the same fixed source  $W_1$  ranges from maximal (Haiku/Sonnet, DeepSeek: unguessable) to largely pre-priced-in (GPT-5.5: 75% guessed), and the ordering tracks vendor training rather than capability, DeepSeek sits with Claude despite being a different vendor, and Opus (the strongest Anthropic model) guesses more than Sonnet.

## 7.5 Generality and controls (summary)

Five follow-up experiments address four threat axes (source-pair generality, the agentic-harness confound, the context-length confound, and prompt-curability of the collapse); full protocols and numbers are in Appendix B. **(i) A second incomparable pair, different domain:** a private **cache**-module contract versus a key-normalization invariant is again certified incomparable on Haiku ( $L_D = +0.774/ +0.496$ ), with one-directional dominance fully certified at  $n = 80$ ; its interference is certified in sign ( $\Psi_T < 0$ , upper bound  $-0.21$ ) but not at the  $\tau = 0.30$  margin, because the 80%  $V_1$  ceiling keeps the four-term bound wide. **(ii) Single-shot replication:** rerunning Haiku in the non-agentic regime preserves certified incomparability ( $L_D = +0.762$  both directions), the trap mis-rank (90% vs 0%), dominance, and one certified interference cell ( $\Psi_T = -0.93$ , upper  $-0.56$ ), so the order-theoretic findings are not agentic-harness artifacts. **(iii) Length-matched padding:** filler character-matched to  $W_{1+}$  leaves  $W_1$  at ceiling (**api\_argorder** 100% vs  $W_{1+}$  10%; interference certified, upper  $-0.41$ ), attributing the collapse to the added signal, not token count. **(iv) Routing instruction:** a one-line convention-routing note recovers most of the collapse (2%  $\rightarrow$  80%, 22%  $\rightarrow$  68%; both separated from  $W_{1+}$ ) while remaining directionally below  $W_1$  alone on the carrying cell; writing that note requires knowing which source is decisive for which task family, which is exactly the coordinate structure the partial order formalizes, so the cure presupposes the diagnosis. **(v) Structured context:** wrapping the same two conventions in labeled XML sections *without* any routing statement mitigates only weakly (2%  $\rightarrow$  42% on the carrying cell, versus 80% with the routing note and 100% for  $W_1$  alone; certified against the raw superset, still far below  $W_1$ ), so segmentation alone does not cure the interference, the operative variable is routing knowledge. The mitigation hierarchy, raw superset  $<$  structure  $<$  routing  $<$  dropping the dominated signal, is itself a finding: it locates the failure in convention-to-task routing rather than in convention segmentation.

## 8 Discussion

With the evidence in hand, we step back to what it licenses. Our results recast a piece of practitioner folklore, “relevance does not transfer across tasks”, as a structural fact about context sources. When two sources are Blackwell-incomparable, the order over them is genuinely partial, and any scalar score, by imposing a total order, must mis-rank them on some task. The empirical program does

not merely illustrate this; it instantiates a single hand-built incomparable pair (a private API-call contract  $W_1$  versus a private wire-encoding invariant  $W_2$  on the same `ledger` module) and shows that the resulting double crossover, the relevance mis-rank, and an anti-monotonicity collapse all reproduce across six models and four vendors. We discuss what the evidence does and does not license, and where the formalization is a model rather than a proof of the realized object.

## 8.1 Scope of the claims

Three scope statements govern everything above. **(i) Two registers.** The theorems concern the idealized Bayes-optimal user of a source; every measured quantity is PASS at one fixed, sub-optimal rule  $\delta_M$ . Lemma 2 supplies the direction the program needs, certified  $(M, \mathcal{D})$ -incomparability on separated decisive coordinates with capped deprived-side value implies Blackwell incomparability, and the converse requires the value-attainment hypothesis (Assumption 1); both hypotheses are enforced by the task design and checked against the measured floors; we therefore read the contribution as a  $\delta_M$ -restricted,  $\mathcal{D}$ -restricted incomparability statement that *earns* the structural conclusion via the bridge rather than assuming it. **(ii) The estimator’s name.**  $\hat{\delta}_{\mathcal{D}}$  performs no garbling minimization and ranges over a fixed finite battery; it is a one-sided, decision-restricted *value-deficiency surrogate*, not a Le Cam deficiency, and no two-sided inequality relates the two for a fixed sub-optimal rule (Remark 4). What is rigorous is the certificate, which we regard as the most robust contribution and which survives independently of the deficiency analogy. **(iii) What the certificate covers.** Its correctness rests on paying the  $\sup_T$  multiplicity via exact intervals and a union bound; its effective support is small (one API cell and one encoding cell carry pair-1 incomparability), each certified claim is controlled *marginally* at  $\eta$  rather than jointly across claims (§6.5), and we describe  $\mathcal{D}$  as a fixed adversarial battery, not a task distribution.

## 8.2 Anti-monotonicity is real, certified, but not capability-ordered

The interference object  $\Psi_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T) - \text{PASS}(W_2, T) + \text{PASS}(\text{none}, T)$  measures whether appending the (irrelevant-to-this-task)  $W_2$  to a sufficient  $W_1$  harms performance. We certify  $\Psi_T \leq -0.30$  on at least one cell for five of six models (Table 4), with collapses as stark as  $100\% \rightarrow 0\%$  ( $\Psi_T = -1.00$ , upper bound  $-0.69$ ) on Sonnet and Kimi. This is a certified empirical falsification of superset-dominance for these fixed rules (Theorem 3): a literal superset of a sufficient signal can sharply reduce PASS, which no monotone-submodular scalar context-value score can represent.

Two corrections follow from the data. **(i) The harm is not monotone in capability.** An early two-point read (Haiku then Sonnet) suggested the collapse “intensifies with model strength”; adding the third Anthropic model (Opus) falsified this. On `api_post_ok` the collapse is  $98 \rightarrow 24\%$  (Haiku),  $100 \rightarrow 0\%$  (Sonnet),  $100 \rightarrow 25\%$  (Opus), and on `api_argorder`  $100 \rightarrow 1\%$ ,  $100 \rightarrow 0\%$ ,  $100 \rightarrow 5\%$  respectively, the middle model (Sonnet) is the starkest and the strongest (Opus) is less so. The defensible claim is robustness across the capability ladder, *not* a capability trend. **(ii) The interference certificate is structurally cell-selected.** Its upper bound includes  $+\text{CP}_{\text{hi}}(p_{\text{none}})$ , so it can only certify harm on cells whose none-baseline floors near zero. This is exactly the set of cells that are unguessable for the model under test, which couples the certifiable-harm set to the model-relative value of private context (§8.3). We treat this as a protocol (Remark 5): we certify anti-monotonicity only on floor-baseline cells, because a leaky baseline mechanically shrinks the certifiable  $\Psi_T$ . On `api_argorder` the harm is certified for Haiku (hardened baseline 1%) and Sonnet (0%) but not for GPT-5.5 (baseline 75%), Kimi (45%), or Opus (15%).

**Will scale restore Blackwell coherence?** The capability non-monotonicity raises a question this framework makes precisely testable: is Blackwell coherence of the realized rule an emergent property of scale? Three points on one vendor’s ladder cannot estimate that curve, and we decline to extrapolate from them; what the framework contributes is the *instrument*: coherence is now a certifiable property of any fixed rule, so “ $\Psi_T$  versus scale” is a measurable trajectory across model generations rather than a speculation, and the present data offer no comfort that the frontier is already there (GPT-5.5 certifies a  $100\% \rightarrow 0\%$  collapse; Opus reproduces the harm at strong point estimates). Two things would remain true under any scaling outcome. First, the theory treats  $\delta_M$  as fixed *per measurement*, not as a claim that models are static: every new model or configuration is a new  $\theta$  (§3.3), and every certified quantity is indexed by it, so model evolution is inside the framework, not outside it. Second, and decisively: anti-monotonicity is a coherence failure of the *rule* and could in principle vanish with scale, but incomparability is a structural property of the *sources*, mutually non-refining signals stay incomparable no matter how capable the reader becomes, so the paper’s central claim, that selection is a partial-order problem, is invariant to how the scaling question resolves; scale can change only how gracefully a fixed rule degrades, never the order itself.

### 8.3 The value of private context is model-relative and tracks vendor training

The value of a private convention,  $I(S; W \mid \theta)$ , is itself a model-relative ( $\theta$ -dependent) object, and the data give a clean continuous map of it. The none-arm baseline on `api_argorder`, how often a model reproduces the private convention with no context supplied, forms a spectrum: Haiku and Sonnet  $\sim 0\%$ , DeepSeek 2%, Opus 15%, Kimi 45%, GPT-5.5 75%. We do not claim this tracks capability. The decisive piece of evidence is DeepSeek: it has a tight, near-zero prior like Claude despite being a different vendor, while GPT-5.5’s prior is wide (55%–75% guessing on `api_argorder/enc_amount`). A “stronger-model-knows-more” account predicts the spectrum to follow capability; it does not. We read  $I(S; W \mid \theta)$  as tracking a vendor’s training distribution rather than raw capability, which makes the theory’s  $\theta$ -dependence concrete and caps which cells certify per model.

### 8.4 Mechanism: the runtime fingerprint is a correlate, not an explanation

Every collapse is accompanied by output-token inflation on the harmful arm:  $W_{1+}$  emits between +87% and +281% more output tokens than  $W_1$  on the API collapse cells (e.g. Haiku +106%/ +87%/ +252%; Sonnet +120%/ +131%; GPT-5.5 +281%/ +274%). One might read this as “verbose thrashing” causing the failure, but the relationship is correlational and uncontrolled: the same superset arm also inflates output by +12% on the *still-passing* encoding cell, which undercuts a clean “verbosity causes collapse” account. We therefore treat the token fingerprint as a reproducible correlate of the harmful arm, not a mechanism.

Two points bound what this paper can and cannot say about mechanism. First, the claim structure is deliberately mechanism-agnostic: Theorem 3 certifies that the realized rule is not Blackwell-coherent *whatever* the internal cause, and the behavioral controls already constrain the hypothesis space from the outside, the harm is not position (order control), not token count (padding control), and is largely, though not fully, curable by explicit convention routing (instruction ablation), a signature consistent with convention interference or mis-binding rather than capacity exhaustion. Second, an activation-level account is genuinely out of reach for the objects studied here: four of the six models are closed-weight production APIs, and the phenomenon is cross-vendor, so any internal story must generalize across architectures we cannot open. The direct path to mechanism is to reproduce the collapse on an open-weight model with the same harness (the launchers speak any OpenAI-compatible endpoint) and then instrument attention and activation patterns on the



contrasting arms; we regard this as the natural mechanistic companion study, downstream of, not prerequisite to, the certified behavioral claim.

**Levels of description, and three readings of the phenomenon.** The partial order is a decision-theoretic, behavioral-level frame; it does not compete with a computational account of how conditioning on added text changes the model’s processing. The two are levels of description of the same event, and the certified behavioral facts are the constraints any computational account must satisfy. Three interpretive readings are compatible with the present data, and each is discriminable by a named experiment. **(i) A transient failure of current models:** discriminated by the coherence-versus-scale trajectory of §8.2, if  $\Psi_T$  shrinks toward zero across model generations, the anti-monotonicity half of the phenomenon was a passing artifact (the incomparability half, being a property of the sources, remains). **(ii) A property of fixed-compute inference:** a rule that spends the same forward computation regardless of context configuration cannot be optimal for every configuration, on this reading some coherence violation persists at every scale and only relocates. **(iii) Prompts as programs:** the added convention does not merely add information, it selects a different computation; this reading is supported by the routing result, a one-line control statement that adds no task-relevant information recovers most of the collapse, and by the structured-context ablation, labeled XML segmentation of the same content recovers only a minority of the collapse ( $2\% \rightarrow 42\%$ ) while the routing statement recovers most of it ( $\rightarrow 80\%$ ), locating the operative variable in convention-to-task *routing* rather than formatting or segmentation. We state these as interpretations, not findings; every certified claim in the paper is independent of which reading is true.

## 8.5 Limitations and threats to validity

We collect the scope restrictions here; each is stated once and sized to its weight.

**From diagnosis to selection.** The constructive selection rule the partial order implies, with its guarantees and evaluation against strong reranker baselines, is factored into a companion paper. Building the certified order on  $m$  candidates costs  $m \cdot |\mathcal{D}| \cdot n$  verifier-gated calls, curation scale, not corpus scale; scaling certified selection to retrieval over  $10^6$  passages, by amortizing the probe into a learned deficiency predictor or running the partial-order step as a re-ranking head over a scalar shortlist, is the central open problem this work opens.

**Source-pair generality.** Incomparability and dominance hold for two hand-built pairs in two domains; the cross-vendor record is single-pair. Pair-2 interference is certified in sign ( $\Psi_T < 0$  at  $n = 80$ ) but not at the  $\tau = 0.30$  margin, its 80%  $V_1$  ceiling keeps the four-term bound wide; pair-2’s relevance trap was an honest null (its prompt was not lexically  $V_2$ -leaning). A third domain and a vendor-general strong mis-rank remain inexpensive next steps.

**Vendor is confounded with harness.** Anthropic models ran agentially, the rest single-shot, so cross-vendor *magnitude* comparisons (the value spectrum, the  $\Psi_T$  ladder) inherit the confound. The binary within-model claims do not, and the single-shot Haiku replication preserves all order-theoretic findings, so they are not agentic artifacts; magnitude comparisons should still be read as suggestive.

**The model is an idealization, twice.**  $\hat{\delta}_{\mathcal{D}}$  certifies dominance and incomparability for a given model on a fixed battery, not the universal Blackwell order. And the projection construction

models sources as clean projections onto independent coordinates; real chunks overlap, contradict, and leak (the measured  $\theta$ -dependence is exactly such leakage), so the construction is the cleanest *sufficient* condition for incomparability, realized exactly by our hand-built pairs, and measuring certified-incomparability prevalence on natural document pairs is the direct next experiment.

**Sampling and certification regime.** The certificate assumes  $n$  i.i.d. Bernoulli PASS draws per cell (Assumption 2); the assumption is most delicate for the agentic arm, where each draw is a fresh multi-turn trajectory (protocol in Appendix C). Interference is certifiable only on cells whose **none** baseline floors near zero, stated as a protocol (Remark 5); marginal-versus-joint control is discussed in §8.1.

**Ranker spectrum and synergy.** The trap mis-rank spans lexical cosine and three representative open-weight learned rankers, with an LLM listwise reranker escaping as predicted, but the learned checkpoints are the small members of their families and the evidence is one trap in one domain. Positive interference (synergy) is untested by construction:  $W_1, W_2$  are task-disjoint, so a complementary pair would be needed to test whether a superset can also *help* beyond its parts; we make no claim about it.

## 9 Conclusion

Context selection for language models is, structurally, a partial-order problem rather than a scalar one. When two context sources are Blackwell-incomparable, no scalar score can rank them correctly for all tasks (Theorem 2), which gives a formal account of the field’s “relevance does not transfer” folklore. We made the order measurable with a decision-restricted value-deficiency surrogate  $\hat{\delta}_{\mathcal{D}}$  equipped with a distribution-free uniform Clopper–Pearson/Bonferroni certificate, and we instantiated the theory on six models across four vendors. On every model we certify ( $\geq 90\%$ ) that two private context sources are empirically incomparable; on every model a real query-conditioned lexical relevance score mis-ranks them; and on five of six models we certify a superset of a sufficient source sharply degrading PASS (a collapse as deep as  $100\% \rightarrow 0\%$ ), a certified falsification of superset-monotonicity for these fixed rules. The value of private context  $I(S; W \mid \theta)$  is itself model-relative, forming a continuous 0%–75% spectrum that tracks vendor training rather than raw capability.

On scope: the theorems concern the idealized Bayes-optimal user of a source; the realized crossover, anti-monotonicity, and value-of-private-context are properties of *fixed* models measured directly, bridged to the structural claim by Lemma 2. The estimator is a regret-gap surrogate, not a Le Cam deficiency; the certificate, which we regard as the most robust contribution, is correct and survives that re-labeling. The evidence now spans two hand-built source pairs in two domains, a single-shot replication that preserves the order-theoretic claims, and a length-matched padding control that attributes the anti-monotonicity collapse to signal content rather than added tokens; the cross-vendor record remains single-pair, vendor is confounded with harness for the magnitude comparisons, and synergy is untested. The practical reading is modest and actionable: selection should be treated as partial-order-aware rather than scalar top- $k$ , the right operational object is a certified deficiency surrogate rather than a relevance scalar, and “the best context” should be expected to be model-relative and only partially ordered. Stated at its sharpest, the impossibility is a theorem about *cheap* selection: per-source scalars (Theorem 2) and topical query-conditioned scores (Proposition 2), not all context selection. The measured spectrum realizes exactly that boundary: lexical cosine, two dense bi-encoders, and a cross-encoder all fall for the trap, while an LLM listwise reranker, which pays per-candidate computation to read the decisive content, escapes

it. The hierarchy is computational: scalar scores cannot represent the order, and probes that read content can measure it, at probe cost. The remaining strengthenings, a vendor-general strong mis-rank, larger reranker checkpoints, and a third domain, are the natural next steps, and the experimental record indicates each is cheap to run.

## References

- Deniz Akdemir. Le Cam distortion: A decision-theoretic framework for robust transfer learning. *arXiv preprint arXiv:2512.23617*, 2025.
- David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.
- Carlo Emilio Bonferroni. Teoria statistica delle classi e calcolo delle probabilità. *Pubblicazioni del R. Istituto Superiore di Scienze Economiche e Commerciali di Firenze*, 8:3–62, 1936.
- Sam Bowyer, Laurence Aitchison, and Desi R. Ivanova. Position: Don’t use the CLT in LLM evals with fewer than a few hundred datapoints. *arXiv preprint arXiv:2503.01747*, 2025. ICML 2025.
- William F. Bradley. Enhancing LLM evaluations: The garbling trick. *arXiv preprint arXiv:2411.01533*, 2024.
- Jaime Carbonell and Jade Goldstein. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336, 1998. doi: 10.1145/290941.291025.
- Kwan Ho Ryan Chan, Yuyan Ge, Edgar Dobriban, Hamed Hassani, and René Vidal. Conformal information pursuit for interactively guiding large language models. *arXiv preprint arXiv:2507.03279*, 2025.
- Deepro Choudhury, Sinead Williamson, Adam Goliński, Ning Miao, Freddie Bickford Smith, Michael Kirchhof, Yizhe Zhang, and Tom Rainforth. BED-LLM: Intelligent information gathering with LLMs and bayesian experimental design. *arXiv preprint arXiv:2508.21184*, 2025. ICLR 2026.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
- Florin Cuconasu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonellotto, and Fabrizio Silvestri. The power of noise: Redefining retrieval for RAG systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.
- Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A. Huerta, and Hao Peng. Context length alone hurts LLM performance despite perfect retrieval. *arXiv preprint arXiv:2510.05381*, 2025. Findings of EMNLP 2025.
- A. Dvoretzky, J. Kiefer, and J. Wolfowitz. Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator. *The Annals of Mathematical Statistics*, 27(3): 642–669, 1956. doi: 10.1214/aoms/1177728174.

- Kawin Ethayarajh, Yejin Choi, and Swabha Swayamdipta. Understanding dataset difficulty with  $\mathcal{V}$ -usable information. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *PMLR*, pages 5988–6008, 2022.
- Loïc Fosse, Frederic Bechet, Benoit Favre, Géraldine Damnati, Gwénolé Lecorvé, Maxime Darrin, Philippe Formont, and Pablo Piantanida. Statistical deficiency for task inclusion estimation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Volume 1: Long Papers*, 2025.
- Norbert Fuhr. A probability ranking principle for interactive information retrieval. *Information Retrieval*, 11(3):251–265, 2008. doi: 10.1007/s10791-008-9045-0.
- Irena Gao, Percy Liang, and Carlos Guestrin. Model equality testing: Which model is this API serving? In *International Conference on Learning Representations (ICLR)*, 2025.
- Kelly Hong, Anton Troynikov, and Jeff Huber. Context rot: How increasing input tokens impacts LLM performance. Technical report, Chroma, 2025. URL <https://www.trychroma.com/research/context-rot>.
- Hai Huang. Directed information  $\gamma$ -covering: An information-theoretic framework for context engineering. *arXiv preprint arXiv:2510.00079*, 2025.
- Bowen Jin, Jinsung Yoon, Jiawei Han, and Sercan O. Arik. Long-context LLMs meet RAG: Overcoming challenges for long inputs in RAG. *arXiv preprint arXiv:2410.05983*, 2024.
- Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. Sufficient context: A new lens on retrieval augmented generation systems. *arXiv preprint arXiv:2411.06037*, 2024. ICLR 2025.
- Lucien Le Cam. Sufficiency and approximate sufficiency. *The Annals of Mathematical Statistics*, 35(4):1419–1455, 1964. doi: 10.1214/aoms/1177700372.
- Lucien Le Cam. *Asymptotic Methods in Statistical Decision Theory*. Springer Series in Statistics. Springer, New York, 1986. doi: 10.1007/978-1-4612-4946-7.
- Will LeVine and Bijan Varjavand. Relevance isn’t all you need: Scaling RAG systems with inference-time compute via multi-criteria reranking. *arXiv preprint arXiv:2504.07104*, 2025. ICLR 2025.
- Shahar Levy, Nir Mazon, Lihi Shalmon, Michael Hassid, and Gabriel Stanovsky. More documents, same length: Isolating the challenge of multiple documents in RAG. *arXiv preprint arXiv:2503.04388*, 2025.
- Ming Liu. More is not always better: Cross-component interference in LLM agent scaffolding. *arXiv preprint arXiv:2605.05716*, 2026.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.
- Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*, 2024.

- Napoleon Paxton. The theorems of dr. david blackwell and their contributions to artificial intelligence. *arXiv preprint arXiv:2604.06621*, 2026.
- Jingxi Qiu, Zeyu Han, and Cheng Huang. SURE-RAG: Sufficiency and uncertainty-aware evidence verification for selective retrieval-augmented generation. *arXiv preprint arXiv:2605.03534*, 2026.
- Stephen E. Robertson. The probability ranking principle in IR. *Journal of Documentation*, 33(4): 294–304, 1977. doi: 10.1108/eb026647.
- Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. Is ChatGPT good at search? Investigating large language models as re-ranking agents. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Erik N. Torgersen. *Comparison of Statistical Experiments*, volume 36 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1991. ISBN 9780521250306. doi: 10.1017/CBO9780511666353.
- Brendan van Rooyen and Robert C. Williamson. Le Cam meets LeCun: Deficiency and generic feature learning. *arXiv preprint arXiv:1402.4884*, 2014.
- M. S. Vinay, Minh-Hao Van, and Xintao Wu. In-context learning demonstration selection via influence analysis. *arXiv preprint arXiv:2402.11750*, 2024.
- Shan Xie, Man Luo, Chadly Daniel Stern, Mengnan Du, and Lu Cheng. DemoShapley: Valuation of demonstrations for in-context learning. *arXiv preprint arXiv:2410.07523*, 2024.
- Jiacheng Ye, Zhiyong Wu, Jiangtao Feng, Tao Yu, and Lingpeng Kong. Compositional exemplars for in-context learning. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- Yue Yu, Wei Ping, Zihan Liu, Boxin Wang, Jiaxuan You, Chao Zhang, Mohammad Shoeybi, and Bryan Catanzaro. RankRAG: Unifying context ranking with retrieval-augmented generation in LLMs. *arXiv preprint arXiv:2407.02485*, 2024. NeurIPS 2024.
- Xiaowei Yuan, Zhao Yang, Ziyang Huang, Yequan Wang, Siqi Fan, Yiming Ju, Jun Zhao, and Kang Liu. Exploiting contextual knowledge in LLMs through V-usable information based layer enhancement. *arXiv preprint arXiv:2504.15630*, 2025. ACL 2025.
- Guang Zhao, Byung-Jun Yoon, Gilchan Park, Shantenu Jha, Shinjae Yoo, and Xiaoning Qian. Pareto prompt optimization. In *International Conference on Learning Representations (ICLR)*, 2025.
- Shuran Zheng, Xuan Qi, Rui Ray Chen, Yongchan Kwon, and James Zou. Proper dataset valuation by pointwise mutual information. *arXiv preprint arXiv:2405.18253*, 2024.

## A Deferred proofs and technical material

### Proof of Theorem 1

*Proof.* This is Blackwell’s theorem [Blackwell, 1953]; the loss-function form and the role of randomization are standard [Le Cam, 1986, Torgersen, 1991]. We use it as a black box. The only substantive specialization is the reading of “decision problem” as “task with a bounded verifier utility” and “experiment” as “context source” (Definition 1), which is faithful because  $v_T \in \{0, 1\}$  is a bounded utility and the kernels of Definition 3 are standard Borel.  $\square$

### Proof of Lemma 1

*Proof.* (a) For deterministic experiments a garbling  $G$  with  $\kappa_{W_B} = G \circ \kappa_{W_A}$  must compute  $\pi_B(S)$  from  $\pi_A(S)$  alone (no further access to  $S$ ). Such a measurable map exists iff  $\pi_B$  is  $\sigma(\pi_A)$ -measurable, i.e.  $\mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A}$ . Under coordinate independence and non-degeneracy,  $\sigma(\pi_B) \subseteq \sigma(\pi_A)$  holds iff every coordinate in  $B$  is among those in  $A$ , i.e.  $B \subseteq A$ : if some  $j \in B \setminus A$ , then  $S_j$  is independent of  $\pi_A(S)$  and non-constant, so it is not  $\sigma(\pi_A)$ -measurable, and no garbling can reconstruct it; conversely if  $B \subseteq A$  the coordinate-deletion map is a (deterministic) garbling. (b) is the contrapositive of (a) applied in both directions.  $\square$

### Proof of Lemma 2

*Proof.* (a) Suppose, for contradiction,  $W_2 \succeq_B W_1$ . By Theorem 1,  $V^*(W_2, T) \geq V^*(W_1, T)$  for every task  $T$ , in particular at  $T_a$ . But  $V^*(W_1, T_a) \geq \text{PASS}_M(W_1, T_a) > g_a = V^*(W_2, T_a)$ , the first inequality because the optimal rule weakly improves on any fixed rule, the second by hypothesis, a contradiction. Hence  $W_2 \not\succeq_B W_1$ . Symmetrically,  $\text{PASS}_M(W_2, T_b) > g_b = V^*(W_1, T_b)$  forbids  $W_1 \succeq_B W_2$ . Neither dominance direction holds, so  $W_1, W_2$  are Blackwell-incomparable. The caps are essential: without them a  $\text{PASS}_M$  crossover is compatible even with Blackwell *equivalence*, since a fixed rule keyed to one signal’s surface form can under-perform on the other, exactly the non-coherence phenomenon Theorem 3 certifies for  $(W_1, W_{1+})$ .

(b) Assumption 1 directly yields  $\text{PASS}_M(W_1, T_a) > \text{PASS}_M(W_2, T_a)$  and the reverse at  $T_b$ , which is Definition 5’s incomparability; the failure of the unconditional converse is witnessed by any fixed rule that violates monotonicity the optimal value obeys.  $\square$

### Proof of Proposition 1

*Proof.* Let  $G$  be the garbling with  $\kappa_{W_2} = G \circ \kappa_{W_1}$ . Any predictor  $g \in \mathcal{V}$  acting on  $W_2$ ’s signal is realized on  $W_1$ ’s signal by the composite  $g \circ G$ , which lies in  $\mathcal{V}$  by the closure hypothesis. Therefore the best  $\mathcal{V}$ -predictor under  $W_1$  attains at least the predictive accuracy of the best under  $W_2$ , so its expected predictive risk is no larger; the risk-reduction relative to the common null signal is thus at least as large under  $W_1$ . The scalarity is immediate:  $I_V$  returns a single real number per source, so it induces a total order and, by Theorem 2, cannot rank a Blackwell-incomparable pair correctly for all tasks.  $\square$

### Proof of Proposition 3

*Proof.* Let  $E$  be the event that all  $2|\mathcal{D}|$  Clopper–Pearson intervals contain their true proportions. Each interval fails with probability at most  $\alpha = \eta/(2|\mathcal{D}|)$  (exact coverage of Clopper–Pearson [Clopper and Pearson, 1934]), so by the union bound  $\Pr[E^c] \leq 2|\mathcal{D}| \cdot \alpha = \eta$ , giving  $\Pr[E] \geq 1 - \eta$ . On  $E$ , for every  $T$  we have  $p_{W_2, T} \geq L_{W_2, T}$  and  $p_{W_1, T} \leq U_{W_1, T}$ , so  $p_{W_2, T} - p_{W_1, T} \geq L_{W_2, T} - U_{W_1, T}$ ; taking

$\max_T$  on both sides,  $\sup_T [p_{W_2,T} - p_{W_1,T}] \geq L_{\mathcal{D}}(W_1, W_2)$ . Thus on  $E$  (probability  $\geq 1 - \eta$ ) the true sup is at least  $L_{\mathcal{D}}(W_1, W_2)$ , which is the claimed lower confidence bound; if it is positive, the true sup is positive, certifying  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ . The reverse direction uses  $L_{W_1,T}$  and  $U_{W_2,T}$ , which are endpoints of the *same*  $2|\mathcal{D}|$  intervals already included in  $E$ ; therefore both one-sided certificates are valid on the single event  $E$ , and their conjunction holds at  $\geq 1 - \eta$  without a further Bonferroni factor.  $\square$

## Proof of Proposition 4

*Proof.* The four arms are independent Bernoulli samples (separate completions), so on the intersection of the four Clopper–Pearson events, probability  $\geq 1 - 4\alpha = 1 - \eta$  by the union bound, each true proportion lies in its interval. Substituting the coverage-respecting endpoint for each term of (6) (upper endpoints for the positively-signed  $p_{W_{1+}}, p_{\text{none}}$ , lower endpoints for the negatively-signed  $p_{W_1}, p_{W_2}$ ) gives  $\Psi_{T^*} \leq \bar{\Psi}_{T^*}$  on that event. For a post-hoc selected cell, replacing  $\alpha$  by  $\eta/(4|\mathcal{D}|)$  and union-bounding over the  $4|\mathcal{D}|$  endpoints preserves  $\geq 1 - \eta$  simultaneous coverage.  $\square$

## Proof of Theorem 3

*Proof.*  $W_{1+} \succeq_B W_1$  by deletion of the extra coordinate (a deterministic garbling, Lemma 1(a)). By Theorem 1 any rule whose value respects  $\succeq_B$  weakly improves under the dominating experiment. The hypothesized strict decrease  $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$  contradicts monotonicity, so  $\delta_M$  is not Blackwell-coherent. A monotone-submodular scalar  $\phi$  assigns  $\phi(W_{1+}) \geq \phi(W_1)$  (monotonicity), hence orders  $W_{1+}$  above  $W_1$  and cannot reproduce the realized reversal.  $\square$

## Why a naive DKW bound fails

*Remark 6* (Why a naive DKW bound fails). The Dvoretzky–Kiefer–Wolfowitz inequality [Dvoretzky et al., 1956] bounds the sup-deviation of *one* empirical CDF from *its own* population CDF over a single i.i.d. sample. Our object is different in two ways that DKW does not handle: it is a max over  $|\mathcal{D}|$  *differences of two binomial means across distinct cells*, and the max *selects* the argmax-gap cell. A single-sample sup inequality neither spans distinct cells nor pays for the selection, so its nominal level under-covers a supremum over  $|\mathcal{D}|$  comparisons; the Type-I error grows with  $|\mathcal{D}|$ . The multiplicity must be paid explicitly via the union bound (Proposition 3), not absorbed into a single-sample DKW envelope.

## Relation to LLM-evaluation statistics

*Remark 7* (Relation to LLM-evaluation statistics). The certificate sits in a small but growing statistical-methodology literature for black-box model evaluation: paired-difference designs and clustered standard errors for evals [Miller, 2024], two-sample tests certifying behavioral differences between API-served models [Gao et al., 2025], and the recommendation to prefer exact frequentist intervals over CLT approximations when per-cell  $n$  is below a few hundred [Bowyer et al., 2025], which is precisely our regime ( $n \in [20, 80]$  per cell) and precisely what the exact Clopper–Pearson construction provides.

## Power and the role of $n$

The asymmetric sample sizes in §7 (Opus at  $n = 20$  certifies incomparability but not interference) are principled, not ad hoc. For a stark cell with empirical proportion at an endpoint ( $\hat{p} \in \{0, 1\}$ ), the

Table 5: Per-cell PASS rates (%) for the carrying cells, by arm. Anthropic rows agentic, others single-shot. “–” marks an arm not in that run. Hardened `api_argorder` (Haiku) lowered the `none` baseline from 40% to 1% while leaving  $W_1$  at 100%.

| Model           | Cell                              | none | $W_1$ | $W_2$ | $W_{1+}$ | $W_{1+}^{\text{rev}}$ |
|-----------------|-----------------------------------|------|-------|-------|----------|-----------------------|
| Haiku-4.5       | <code>api_post_ok</code>          | 0    | 98    | 0     | 24       | 1                     |
| Haiku-4.5       | <code>api_argorder</code> (hard.) | 1    | 100   | 0     | 1        | 44                    |
| Haiku-4.5       | <code>enc_amount</code>           | 0    | 0     | 100   | –        | –                     |
| Haiku-4.5       | <code>trap_store_wire</code>      | 0    | 96    | 0     | –        | –                     |
| Sonnet-4.6      | <code>api_post_ok</code>          | 0    | 100   | 0     | 0        | –                     |
| Sonnet-4.6      | <code>api_argorder</code>         | 0    | 100   | 0     | 0        | –                     |
| Sonnet-4.6      | <code>trap_store_wire</code>      | 0    | 100   | 0     | –        | –                     |
| Opus-4.8        | <code>api_post_ok</code>          | 0    | 100   | 0     | 25       | –                     |
| Opus-4.8        | <code>api_argorder</code>         | 15   | 100   | 0     | 5        | –                     |
| Opus-4.8        | <code>trap_store_wire</code>      | 15   | 100   | 10    | –        | –                     |
| GPT-5.5         | <code>api_post_ok</code>          | 5    | 100   | 0     | 0        | –                     |
| GPT-5.5         | <code>api_argorder</code>         | 75   | –     | –     | –        | –                     |
| GPT-5.5         | <code>enc_amount</code>           | 55   | –     | –     | –        | –                     |
| GPT-5.5         | <code>trap_store_wire</code>      | 0    | 100   | 0     | –        | –                     |
| DeepSeek-V4-Pro | <code>api_post_ok</code>          | 0    | 75    | 0     | 0        | –                     |
| DeepSeek-V4-Pro | <code>api_argorder</code>         | 2    | 68    | 0     | –        | –                     |
| DeepSeek-V4-Pro | <code>trap_store_wire</code>      | 0    | 98    | 0     | –        | –                     |
| Kimi-K2.6       | <code>api_post_ok</code>          | 0    | 100   | 0     | 0        | –                     |
| Kimi-K2.6       | <code>api_argorder</code>         | 45   | –     | –     | –        | –                     |
| Kimi-K2.6       | <code>trap_store_wire</code>      | 0    | 100   | 0     | –        | –                     |

Clopper–Pearson interval collapses to a one-sided bound that clears 0 at small  $n$ : the incomparability certificate (5) is carried by such stark cells ( $W_1$  at  $\approx 100\%$ ,  $W_2$  at  $0\%$  and vice versa), so  $n = 20$  suffices. The interference certificate (7) mixes a possibly-intermediate `none` baseline and the  $W_{1+}$  arm; with non-endpoint proportions the intervals are wide and  $n = 20$  is underpowered. The two certificates therefore have genuinely different effective- $n$  requirements, which is why the consolidated record marks Opus incomparability as certified and Opus interference as reproduced but not certified.

## B Extended experimental record

### Generality and controls: full protocols

Five follow-up experiments address four threat axes directly: source-pair generality, the agentic-harness confound, the context-length confound, and whether the anti-monotonicity collapse is curable by cheap prompting.

**A second incomparable pair, different domain.** We repeated the protocol with a second hand-built pair on a private `cache` module:  $V_1$ , the `cache.put(key, value, *, ttl)` call contract (argument order, required keyword-only `ttl`, a True/False “newly inserted” flag, a `CacheError` type), versus  $V_2$ , a key-normalization invariant (a fixed `kx7-` namespace prefix and an underscore rule). On Haiku at  $n = 40$  the pair is again certified incomparable ( $L_{\mathcal{D}} = +0.774/ +0.496$ , both directions clear of zero) with a clean double crossover ( $V_1$  solves the contract and trap cells,  $V_2$



solves key-normalization, all none-baselines floor at 0%), and the dominance control again certifies one-directionally. Anti-monotonicity reproduces in direction at  $n = 40$  (the superset collapses the contract cell  $80\% \rightarrow 28\%$ ,  $\Psi_T = -0.53$ ); re-running the four arms of the contract cell at  $n = 80$  shows the collapse is stable ( $80\% \rightarrow 29\%$ ,  $\Psi_T = -0.51$ ) with Clopper–Pearson upper bound  $-0.21$ , which certifies *strictly negative* interference ( $\Psi_T < 0$  at  $\geq 90\%$ ) but does not clear the pre-registered magnitude margin  $\tau = 0.30$ : the binding constraint is the  $V_1$  ceiling itself ( $80\%$ , versus pair-1’s  $\sim 100\%$ ), which keeps the four-term bound wide regardless of  $n$ . We report pair-2 interference as a certified sign, not a certified magnitude. The  $n = 80$  run also strengthens the dominance control to a fully certified one-directional dominance ( $U_{\mathcal{D}} = 0.000$  with a certified strict win). The relevance trap for this pair was a null, its prompt vocabulary leaned toward  $V_1$  ( $\varphi(V_1) = 0.39 > \varphi(V_2) = 0.25$ ), so relevance *agreed* with PASS, so the strong mis-rank remains a pair-1 result. The contribution of the second pair is therefore *generality of the incomparability and dominance results to a new domain*, not a second mis-rank.

**Single-shot breaks the harness confound.** To separate vendor from harness, we ran an Anthropic model (Haiku) in *single-shot completion* mode, one turn, no tools, code extracted from the returned text, the same regime as the non-Anthropic models. At  $n = 40$  the core results survive the non-agentic harness: incomparability is certified ( $L_{\mathcal{D}} = +0.762$  both directions), the relevance mis-rank reproduces (trap PASS  $W_1$  90% vs  $W_2$  0%), the dominance control certifies, and anti-monotonicity is certified on `api_argorder` ( $\Psi_T = -0.93$ , upper bound  $-0.56$ ). The one regime-dependent quantity is absolute execution reliability: single-shot  $W_1$  rescue on `api_post_ok` is 52% versus 98% agentic (no retries to self-correct), so that cell’s interference does not certify single-shot while its order-theoretic claims do. The order-theoretic findings, certified incomparability, relevance mis-rank, dominance, and at least one certified interference cell, are therefore *not* artifacts of the agentic harness.

**A length-matched padding control.** Anti-monotonicity compares  $W_1$  against the strictly longer  $W_{1+}$ , and added input length alone is known to degrade performance [Du et al., 2025], so the collapse could in principle be a token-count effect rather than a signal effect. To isolate this we added a padding arm  $W_{\text{pad}}$ : filler text in the same “team context (private)” register as  $W_2$ , placed in the same position  $W_2$  occupies in  $W_{1+}$  (padding first,  $W_1$  last), character-matched to  $W_{1+}$  within 13 characters, but carrying no coordinate any verifier checks (office-logistics content). On single-shot Haiku at  $n = 40$  on the two collapse cells, padding is free while the equal-length dominated signal is catastrophic: on `api_argorder`,  $W_1$  98%,  $W_{\text{pad}}$  100%,  $W_{1+}$  10%; on `api_post_ok`,  $W_1$  50%,  $W_{\text{pad}}$  65%,  $W_{1+}$  10%; and the same-run interference certificate clears on `api_argorder` ( $\Psi_T = -0.82$ , upper bound  $-0.41$ ). The anti-monotonicity harm is therefore attributable to  $W_2$ ’s signal content interacting with the fixed rule, not to added tokens. One residual axis is not isolated by this control: the padding is off-topic for the `ledger` module while  $W_2$  is on-topic, so “topical but verifier-irrelevant” text remains a smaller, untested middle case.

**A routing-instruction ablation: the cure presupposes the diagnosis.** If the collapse reflects convention confusion, an explicit routing note might prevent it, and if a one-line prompt fully cured the harm, its practical significance would shrink. We measured this directly: the arm  $W_{1+}^{\text{instr}}$  appends to  $W_{1+}$  a single routing sentence stating which convention serves which task family (“for tasks that call the ledger API use only the API call contract; the wire-encoding invariant applies only to encode/decode tasks”). On single-shot Haiku at  $n = 40$  the instruction recovers most, but not all, of the collapse: on `api_argorder`,  $2\% \rightarrow 80\%$  ( $W_1$  alone: 100%), and on `api_post_ok`,  $22\% \rightarrow 68\%$

( $W_1$  alone: 50%); both recoveries over  $W_{1+}$  are large (non-overlapping exact 95% intervals), while the residual gap below  $W_1$  alone on the carrying cell (80% vs. 100%, intervals  $[0.62, 0.92]$  vs.  $[0.90, 1.00]$ ) is directional but not separated at this  $n$ . We pre-commit to the two readings jointly. First, the harm is substantially mediated by convention interference that explicit routing knowledge prevents, so pipelines that can attach per-task routing labels have a cheap mitigation. Second, and decisively for the paper’s thesis: *writing the routing sentence requires already knowing which source is decisive for which task family*, which is precisely the coordinate structure the partial order formalizes and the deficiency certificate measures; the cure is an injection of selection-time coordinate knowledge into the prompt, not a scalar score, and no query-independent scalar can supply it (Theorem 2). Formally,  $\delta_M$  with the routing note is a *different, more Blackwell-coherent rule*; the certified falsification of Theorem 3 concerns the base rule and is unaffected, and even under the instructed rule, keeping the dominated signal is at best equal and directionally worse than dropping it, so the selection-level conclusion, keep the non-dominated set rather than forcing the superset, survives the ablation intact.

**A structured-context ablation: segmentation is not the bottleneck.** To separate “the model cannot segment the two conventions” from “the model cannot route them to task families,” the arm  $W_{1+}^{\text{xml}}$  wraps the same two conventions, in the same order as  $W_{1+}$ , in labeled XML sections (`<wire_encoding_invariant>`, `<api_call_contract>`) with no routing statement. On single-shot Haiku at  $n = 40$  per cell, structure alone mitigates weakly: `api_argorder` 42% (vs.  $W_{1+}$  2%,  $W_{1+}^{\text{instr}}$  80%,  $W_1$  100%) and `api_post_ok` 25% (vs. 12%, 68%, 57%). The structure-only mitigation is separated from the raw superset on the carrying cell (exact 95% intervals  $[0.27, 0.59]$  vs.  $[0.001, 0.13]$ ) but remains far below  $W_1$  alone; on `api_post_ok` it is not separated from the raw superset. Protocol note: the  $W_{1+}^{\text{xml}}$  arm was measured in a clean arm-only re-run (its arm in the first run was invalidated by a transport-failure window and scored no completions; the re-run had zero transport errors), so its comparison against same-day baseline arms is across runs rather than within one run. The resulting mitigation hierarchy, raw superset  $<$  structure  $<$  routing  $<$  dropping the dominated signal, locates the failure in convention-to-task routing.

## C Reproducibility and protocol

All reported numbers come from real model calls whose outputs are checked by executable verifiers; there are no modeled or aspirational figures. The complete artifact bundle—the measurement harness, the executable verifiers, and every per-model run log and result JSON cited below—is available at <https://github.com/abilous-ti/blackwell-llm-context>. The harness (`tokenbench/measure_blackwell.py`) is Python-standard-library only (no third-party dependencies; the Clopper–Pearson endpoints use a stdlib regularized incomplete beta), is cross-OS, and implements the surrogate  $\hat{\delta}_{\mathcal{D}}$ , the uniform incomparability certificate (`certify_incomparable`, `clopper_pearson`), the dominance control (`certify_dominance`), the interference certificate, the sample-size helpers (`required_n`, `required_n_stark`), and the lexical-relevance scalar (`lexical_relevance`). It includes launchers for the Anthropic CLI (`claude -p`), the Azure Responses API, and OpenAI-compatible chat completions; the launcher auto-detects API style from the endpoint, and all API keys are read from the environment (never written to a file).

**Protocol.** The verifier stubs a fake `ledger` module that accepts *only* the exact private contract (it rejects wrong positional arg order, a missing `memo=` keyword, and the wrong exception type); return code 0 is PASS. The four certified cells are `api_post_ok`, `api_argorder` (hardened so `memo`

arrives via a parameter, forcing the keyword-only fact and dropping its baseline from 40% to 1%), `enc_amount`, and `trap_store_wire`; arms are  $\{\text{none}, W_1, W_2, W_{1+}\}$  plus  $W_{1+}^{\text{rev}} = W_1++W_2$  for the order control and  $W_{\text{pad}}$  (length-matched neutral filler  $++W_1$ ) for the padding control. PASS is measured over  $n$  i.i.d. completions per cell and the certificate uses  $\eta = 0.10$ . The Anthropic arms run agentially (`claude -p`, up to six turns); the non-Anthropic arms run single-shot (one completion + code extraction). We note for transparency that this design choice, agentic for Anthropic, single-shot for the others, is the vendor/harness confound discussed in §8.5.

**Decoding, seeding, retries, and turn budget.** All runs use the provider’s default decoding configuration: the harness sets no explicit temperature, top- $p$ , or seed for any arm (the Claude CLI and the Azure/OpenAI-compatible launchers are invoked without sampling overrides), so the  $n$  draws per cell are independent samples from each provider’s default decoder. Each draw is a fresh process invocation with no state shared across draws; the agentic arm allows up to six turns, the single-shot arms exactly one, with a 200-second timeout per call. Transport-level failures (empty or zero-token results) are retried up to three times with quadratic backoff; a draw that still fails is scored as a failure, never as a pass. Runs recorded before the padding control predate the retry wrapper and used single attempts, under which a transport failure scores as a failure; the stark 98–100%  $W_1$  ceilings in the certified anchors indicate such failures were negligible in those runs.

**Artifacts.** Per-model run logs and result JSONs are in the artifact repository (<https://github.com/abilous-ti/blackwell-llm-context>) under the pattern `blackwell_*_n*.log,json`, including `blackwell_n80_run.log / blackwell_n80_results.json` (the  $n = 80$  Haiku bundled certification, 1600 calls), `blackwell_argorder_n80.json,log` (the hardened second anti-monotonicity cell), `blackwell_sonnet_n40`, `blackwell_opus_n20`, `blackwell_gpt55_n40`, `blackwell_deepseek_n40`, `blackwell_kimi_n40`, `blackwell_pad_n40b` (the length-matched padding control, single-shot, 400 calls), `blackwell_pair2_n80` (the pair-2 contract cell at  $n = 80$ , 320 calls), `blackwell_instr_n40` (the routing-instruction ablation, 400 calls), and `blackwell_xml_n40` and `blackwell_xml_arm_only` (the structured-context ablation; the first run’s XML arm is transport-invalidated and superseded by the arm-only re-run), `reranker_trap_n30` (the LLM listwise reranker probe, 90 calls; harness `tokenbench/measure_reranker_trap.py`), and `dense_trap.json` (the dense/cross-encoder probe, deterministic local scoring of open-weight checkpoints at zero API cost; harness `tokenbench/measure_dense_trap.py`). The full experimental record with exact per-cell figures is in `EXPERIMENT-BLACKWELL.md` (§§10–13) and the theory hardening (formalization and proof sketches) in `BLACKWELL.md` (§10).

**Compute and cost.** Total measured spend was approximately \$210 on Anthropic (`claude -p`): roughly \$123 across the Haiku and Sonnet anchors plus the Opus run (\$49.84), the order-control and second-cell runs, the padding control (\$7.73), the pair-2  $n = 80$  run (\$10.05), the routing-instruction ablation (\$7.90), and the structured-context ablation (\$8.40 across an arm-invalidated first run and a clean arm-only re-run); the dense/cross-encoder probe ran locally on open weights at zero API cost. The non-Anthropic runs (GPT-5.5, DeepSeek, Kimi) used Azure quota; the Azure Responses API returns no cost field, so those rows report a \$0 cost artifact while token counts are captured, the \$210 figure is the Anthropic spend, not the total cost of all six models.

**Reference verification.** The bibliography was assembled in part with automated literature search and every entry was subsequently verified against arXiv/DBLP/publisher records; all entries resolve to real papers, and the classics (Blackwell 1953; Le Cam 1964/1986; Torgersen 1991; Ethayarajh et al. 2022) were checked against the primary sources.